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Data-driven Reduced Order Modelling of Vortex and Buoyancy-driven Incompressible Turbulent Flows in 2D

Bachelor's Thesis

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Abstract

This thesis investigates data-driven reduced-order models for two-dimensional incompressible flows with vortex interaction and buoyancy-driven convection. The main goal is to assess whether the exact operator-inference approach of Rosenberger et al. [29] can, in the two-dimensional incompressible-flow setting considered here, recover intrusive projection-based reduced operators without explicitly assembling the full-order polynomial operators, and whether the resulting reduced models provide useful surrogates for challenging flow problems.

Two test cases are considered: the normal collision of a vortex dipole with a no-slip wall at $\mathrm{Re} = 2500$, and Rayleigh–Bénard convection at $\mathrm{Ra} = 1.6 \times 10^7$ and $\mathrm{Pr} = 1$. Full-order data are generated with an energy-conserving staggered finite-volume solver based on the discretization framework of Agdestein and Sanderse [1], and the POD bases are computed using the same finite-volume inner products as the solver. Exact operator inference is implemented through rank-ensuring one-step full-order queries and compared with matrix-free intrusive Galerkin projection and conventional trajectory-based operator inference.

For both test cases, exact operator inference recovers the corresponding intrusive reduced operators to round-off accuracy. The remaining trajectory errors are therefore mainly caused by POD truncation, regime representation and online approximation choices rather than by operator identification. In the dipole–wall collision, a global POD basis captures the energetic dynamics but struggles with localized near-wall vorticity after impact. In the Rayleigh–Bénard case, a scheduled switched ROM with local pre-reversal, transition and post-reversal bases is required to represent the circulation reversal. The results show that exact operator inference is a reliable accuracy-oriented way to construct intrusive-equivalent affine-quadratic ROMs, while practical surrogate performance depends strongly on the reduced space, retained fluctuation energy and regime coverage of the training data.

Preface

This thesis was written as part of the Combined Bachelor End Project for the BSc degrees in Applied Mathematics and Applied Physics at Eindhoven University of Technology. The project was carried out from February to July 2026. Working on this project gave me the opportunity to combine numerical methods for fluid dynamics with data-driven techniques, reduced-order modelling, and data analysis. In particular, I focused on the simulation of incompressible flow problems and on the construction and assessment of reduced models for representative flow cases. Moreover, I had the chance to deepen my abilities in scientific research and writing, as well as in the use of scientific computing tools.

I would like to thank my supervisors for their guidance and feedback throughout the project. In particular, I would like to thank Dr.ir. Benjamin Sanderse for introducing me to possible projects within computational fluid dynamics and for his guidance during a large part of the project. I would also like to thank Dr.ir. Rudie Kunnen for his adaptability in helping us define a project that suited both physics and mathematics, for the numerous discussions on Rayleigh–Bénard convection, and for his guidance throughout the whole project. Finally, I would like to thank Prof.dr. Karen Veroy-Grepl for undertaking the role of supervisor during the last months of the project and for her feedback and guidance in the final stages. I am also grateful to Henrik Rosenberger for his introduction and support in using his novel approach to non-intrusive reduced-order modelling, and to Syver Agdestein for making available and supporting `IncompressibleNavierStokes.jl`, the Julia incompressible Navier–Stokes solver on which this work builds. Its staggered-grid discretization, pressure-projection solvers, time-stepping routines, and simulation infrastructure formed the numerical foundation used throughout this project.

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Chapter 1

Introduction

Fluid flows have an important place in many systems in nature as well as industrial applications, including atmosphere dynamics, ocean dynamics, industrial aerodynamics, heat transfer processes, and renewable energy systems. The analysis and forecasting of fluid flows are therefore important both in terms of science and industry. In many instances, the mathematical description of fluid motion, in particular, the Navier–Stokes equations for incompressible fluids, is impossible to solve analytically and needs to be computed using numerical methods.

The main subject of this thesis is turbulent incompressible flows in which the flow dynamics are caused by two physical processes: vortex interaction and buoyancy effects. Vortex-driven flows are dominated by the motion, deformation and interaction of coherent vortical structures, for which the *Reynolds number*[1] is a fundamental parameter. A representative example is the collision of a vortex dipole with a no-slip wall. Although this problem appears geometrically simple, Clercx and Bruneau show that the collision produces small-scale vorticity near the wall, strong boundary-layer effects and coherent secondary vortices, making grid-converged full-order simulations difficult to obtain [13].

On the other hand, buoyancy-driven flows occur because of density differences as a result of temperature differences, which cause flow motion due to gravity forces. One well-known case of buoyancy-induced flows is the Rayleigh–Bénard convection. In this phenomenon, flows within a fluid layer heated from its bottom surface and cooled at its top surface exhibit rolling motions, thermal plumes, and even turbulence for sufficiently high *Rayleigh numbers*[2]. The flows have been widely used as a model problem to investigate thermal convection and related physics; refer to the review by Ahlers, Grossmann and Lohse [2].

Simulation of such flows is costly, especially when high resolution is needed to resolve boundary layers, coherent vortices, thermal plumes and heat-transfer diagnostics. This cost becomes a major obstacle when simulations must be repeated many times, for example in parameter studies, optimization, control, uncertainty quantification or digital-twin applications. The need for computational efficiency in such repeated-query settings is one of the main motivations for reduced-order models (ROMs) [18, 30].

Reduced-order models aim to approximate the dominant dynamics of a high-dimensional simulation in a much lower-dimensional space. Instead of evolving the full discretized velocity or temperature fields, the flow is represented through a small number of reduced coordinates associated with the most relevant structures in the data. Proper orthogonal decomposition (POD) is one of the classical tools for extracting energetic coherent structures

[1] The Reynolds number is a dimensionless parameter which compares inertial and viscous effects in a flow. The higher it is, the more turbulent the fluid's behaviour is.

[2] The Rayleigh number is the main control parameter for buoyancy-driven thermal convection, comparing destabilizing buoyancy effects with viscous and thermal diffusion. High Rayleigh numbers indicate stronger buoyancy effects, thinner thermal boundary layers, more intense plume formation and, eventually, chaotic or turbulent convection [2].

from turbulent flow data [5, 36].

The focus of this thesis is specifically on data-driven reduced-order modelling. In contrast to intrusive reduced models, which require direct access to the discrete operators of the full-order solver, data-driven and non-intrusive methods infer the reduced dynamics from simulation data. Operator inference (OpInf) is one such approach: it learns reduced operators from snapshot data while retaining a structure inspired by projection-based ROMs. This makes it particularly attractive when the full-order model (FOM) solver is available as a source of data but its internal operators are difficult to access or manipulate [27].

The aim of this thesis is therefore to investigate whether data-driven ROMs can provide accurate and efficient surrogate descriptions of challenging incompressible flows. By comparing reduced-order predictions with full-order simulations, the thesis assesses to what extent data-driven ROMs can reproduce the dominant physical mechanisms, energetic behaviour and coherent structures of turbulent incompressible flows. A snapshot-generation technique developed by Rosenberger et al. [29] is implemented, as it guarantees exact reconstruction of intrusive projection-based ROM operators under suitable assumptions. Whereas the reference study demonstrates the method on one-dimensional model problems, this thesis applies it to two-dimensional incompressible-flow test cases.

The thesis also considers a regime-local ROM strategy. A single global linear basis can be inefficient when coherent structures translate, rotate, or when the flow changes from one dominant pattern to another; this limitation motivates local and adaptive reduced bases in transport-dominated and convection-dominated problems [25, 17, 3]. In this work the idea is tested on a Rayleigh–Bénard trajectory with a change in the large-scale circulation. Several local ROMs are trained on different time regimes, and the switching times are determined from the normalized angular-momentum history of the reference FOM and then kept fixed during the ROM prediction.

The remainder of this report is organized as follows. Chapter 2 introduces the physical parameters, diagnostic quantities and numerical concepts used in the two test cases. Chapter 3 describes the continuous FOM equations, their discrete polynomial structure and the problem setup used to generate data. Chapter 4 presents POD, intrusive projection, operator inference and the local switching strategy. Chapter 5 reports the dipole–wall and Rayleigh–Bénard numerical experiments. The final chapter summarizes the conclusions and limitations, while additional implementation details are collected in the appendices.

Chapter 2

Theoretical Background

This chapter introduces the main physical and numerical concepts used throughout the thesis. The goal is not to give a complete derivation of fluid mechanics, but to define the dimensionless parameters, approximations and diagnostic quantities needed to understand the dipole–wall collision and Rayleigh–Bénard convection test cases.

2.1 Non-dimensional numbers

A common way to study fluid flows is to compare the relative importance of different physical mechanisms. This is done by introducing characteristic scales and rewriting the governing equations in dimensionless form [39, 2]. The resulting dimensionless numbers are important because they determine which physical effects dominate the flow and allow simulations with different physical units to be compared.

Let L denote a characteristic length scale, U a characteristic velocity scale, ΔT a characteristic temperature difference, ν the kinematic viscosity, κ the thermal diffusivity, g the gravitational acceleration, and α the thermal expansion coefficient. These parameters are the physical scales and material properties that enter the Reynolds, Prandtl and Rayleigh numbers used below [39, 2].

2.1.1 Reynolds number

The Reynolds number is a dimensionless quantity that measures the relative importance of inertial and viscous effects in a fluid flow. With the notation introduced above, it is defined as

$$Re = \frac{UL}{\nu}. \tag{2.1}$$

This interpretation can be understood by comparing the order of magnitude of the different terms in the incompressible Navier–Stokes equations [39]. The convective inertial term scales as

$$\boldsymbol{u} \cdot \boldsymbol{\nabla} \boldsymbol{u} \sim \frac{U^2}{L}, \tag{2.2}$$

whereas the viscous diffusion term scales as

$$\nu \Delta \boldsymbol{u} \sim \nu \frac{U}{L^2}. \tag{2.3}$$

The ratio between inertial and viscous effects is therefore

$$\frac{U^2/L}{\nu U/L^2} = \frac{UL}{\nu} = Re. \tag{2.4}$$

Hence, the Reynolds number quantifies whether the flow dynamics are mainly governed by inertia or by viscous diffusion.

Small Reynolds numbers correspond to flows in which viscous effects dominate. Conversely, large Reynolds numbers correspond to flows in which inertial transport is dominant, while viscosity mainly acts at small scales or near solid boundaries. As a consequence, high Reynolds-number flows may develop thin boundary layers, small vortical structures, and, in sufficiently complex regimes, turbulent behaviour.

2.1.2 Prandtl number

The Prandtl number is defined as

$$Pr = \frac{\nu}{\kappa}. \tag{2.5}$$

It compares momentum diffusion to thermal diffusion. If $Pr \ll 1$, heat diffuses much faster than momentum. If $Pr \gg 1$, momentum diffuses much faster than heat. If $Pr = 1$, both effects occur on comparable time scales.

The Prandtl number is especially important in Rayleigh–Bénard convection because the problem contains both a velocity field and a temperature field. It influences the relative thickness of the velocity and thermal boundary layers and affects the structure of thermal plumes and convective rolls [2]. The Rayleigh–Bénard simulations in this thesis use $Pr = 1$, so momentum and heat diffuse on comparable time scales.

2.1.3 Rayleigh number

The Rayleigh number is the main control parameter in Rayleigh–Bénard convection. For the two-dimensional configuration considered in this thesis, the characteristic length scale is the cavity height H. The lower plate is kept at temperature T_b, the upper plate at temperature T_t, and the imposed temperature difference is

$$\Delta T = T_b - T_t > 0. \tag{2.6}$$

Using the material parameters introduced at the beginning of this section, the Rayleigh number is

$$Ra = \frac{g\alpha\Delta T H^3}{\nu\kappa}, \tag{2.7}$$

so H takes the role of the characteristic length L in this problem.

Physically, Ra compares the destabilizing effect of buoyancy with the stabilizing effects of viscosity and thermal diffusion [41]. Increasing g, α, ΔT, or H increases Ra, while increasing either diffusivity decreases it. Thus, larger values correspond to stronger buoyancy forcing relative to damping by momentum and thermal diffusion. The Rayleigh number can also be written as the product of the Grashof number and the Prandtl number,

$$Ra = Gr\,Pr, \qquad Gr = \frac{g\alpha\Delta T H^3}{\nu^2}. \tag{2.8}$$

This form emphasizes that Ra combines the strength of buoyancy-driven motion, represented by the Grashof number Gr, with the relative importance of momentum and thermal diffusion, represented by the previously defined Prandtl number Pr [41].

For sufficiently small Ra, the purely conductive state is stable: the velocity field remains zero and heat is transported mainly by diffusion. When Ra exceeds a critical value, this conductive state becomes unstable and convection starts [2]. For the classical Rayleigh–Bénard problem with fixed temperatures at the horizontal plates and no-slip velocity boundary conditions, the critical value is approximately

$$Ra_c \approx 1708. \tag{2.9}$$

Above this threshold, the flow develops organized convective structures, such as rolls. As Ra is increased further, the flow may become time-dependent and, eventually, turbulent [2].

2.1.4 Nusselt number

The Nusselt number measures the efficiency of heat transport. It is an essential parameter used for validation of Rayleigh–Bénard convection simulations. It is defined as the ratio between the total vertical heat flux and the heat flux that would occur by pure conduction:

$$Nu = \frac{\text{total vertical heat flux}}{\text{conductive heat flux}}. \tag{2.10}$$

Thus, $Nu = 1$ corresponds to pure conduction, while $Nu > 1$ means that convection enhances heat transport.

For the same two-dimensional Rayleigh–Bénard setting, let $\boldsymbol{u} = (u_x, u_y)^T$ be the velocity field and $T(x, y, t)$ the temperature field. The horizontally averaged instantaneous Nusselt number at height y can be written as

$$Nu(y,t) = \frac{\langle u_y T \rangle_x - \kappa \langle \partial_y T \rangle_x}{\kappa \Delta T / H}, \tag{2.11}$$

where $\langle \cdot \rangle_x$ denotes horizontal averaging. The numerator is the sum of the convective vertical heat flux, carried by $u_y T$, and the conductive vertical heat flux, represented by $-\kappa \partial_y T$. The denominator is the conductive heat flux of the reference state between the two plates.

At the bottom and top no-slip walls, the vertical velocity is zero. Therefore, the wall Nusselt numbers are determined only by the wall-normal temperature gradient:

$$Nu_b(t) = -\frac{H}{\Delta T} \langle \partial_y T(x,0,t) \rangle_x, \qquad Nu_t(t) = -\frac{H}{\Delta T} \langle \partial_y T(x,H,t) \rangle_x. \tag{2.12}$$

For the non-dimensional case $H = 1$ and $\Delta T = 1$, this reduces to

$$Nu_b(t) = -\langle \partial_y T(x,0,t) \rangle_x, \qquad Nu_t(t) = -\langle \partial_y T(x,1,t) \rangle_x. \tag{2.13}$$

In numerical simulations, one may also compute a volume-averaged Nusselt number, which averages the vertical heat flux over the full computational domain:

$$Nu_V(t) = \frac{H}{\kappa \Delta T} \langle u_y T - \kappa \partial_y T \rangle_V, \tag{2.14}$$

where $\langle \cdot \rangle_V$ denotes volume averaging. In the non-dimensional formulation used in the simulations, $H = 1$, $\Delta T = 1$, and the velocity is scaled with κ / H. The volume-averaged expression then becomes

$$Nu_V(t) = \langle u_y T - \partial_y T \rangle_V. \tag{2.15}$$

Because the domain has unit height with $T = 1$ at the bottom plate and $T = 0$ at the top plate, the volume average of $-\partial_y T$ contributes 1. Thus, equivalently,

$$Nu_V(t) = \langle u_y T \rangle_V + 1. \tag{2.16}$$

Volume-averaged Nusselt numbers are commonly used in numerical studies of Rayleigh–Bénard convection, together with the top and bottom wall values [21].

In a statistically steady simulation without internal heat sources, the long-time averaged Nusselt number measured at the bottom wall, top wall, and through the volume should agree up to numerical error:

$$\overline{Nu_b} \approx \overline{Nu_t} \approx \overline{Nu_V}. \tag{2.17}$$

For this reason, the Nusselt number is a standard validation quantity in numerical studies of Rayleigh–Bénard convection [2, 35, 21].

2.1.5 Angular momentum of the large-scale circulation

In addition to heat-transfer diagnostics, Rayleigh–Bénard convection is often interpreted in terms of the large-scale circulation formed by the convective roll [2]. In studies of circulation reversals, global rotation or angular-impulse type quantities are used to track the orientation and strength of this motion [10, 7]. A convenient scalar diagnostic for the present two-dimensional square domain is the angular momentum of the velocity field around the centre of the domain. For a two-dimensional velocity field $\boldsymbol{u} = (u_x, u_y)^T$ on a domain Ω, and for a reference point (x_c, y_c), the out-of-plane angular momentum is

$$L_z(\boldsymbol{u}) = \int_\Omega \left[(x - x_c)u_y(x,y) - (y - y_c)u_x(x,y)\right] d\Omega. \tag{2.18}$$

The sign of L_z indicates the dominant orientation of the circulation. Positive and negative values correspond to opposite rotating states, while values close to zero indicate that the large-scale circulation is weak or that the flow is passing through a reversal.

The magnitude of L_z depends on the kinetic-energy level of the flow. A natural normalization is obtained by analogy with rigid-body rotation, where angular momentum, moment of inertia and rotational kinetic energy are related through the same inertia tensor [38]. For comparing different times during the same simulation, the angular momentum is therefore normalized by the kinetic energy and by the moment of inertia of the domain:

$$\ell_z(\boldsymbol{u}) = \frac{L_z(\boldsymbol{u})}{\sqrt{I_z\, 2E(\boldsymbol{u})}}, \qquad I_z = \int_\Omega \left[(x - x_c)^2 + (y - y_c)^2\right] d\Omega. \tag{2.19}$$

Here $E(\boldsymbol{u})$ is the kinetic energy defined in (2.27). The normalization removes two scale effects from L_z: $\sqrt{2E(\boldsymbol{u})}$ removes the dependence on the overall velocity magnitude, while $\sqrt{I_z}$ removes the geometric scale set by the domain and the chosen centre. The square-root scaling follows from the rigid-body relation $E_{\mathrm{rot}} = L_z^2/(2I_z)$, or equivalently $|L_z| = \sqrt{I_z\, 2E_{\mathrm{rot}}}$ [38]. Thus ℓ_z is dimensionless: its sign gives the circulation direction, and its magnitude indicates how coherently the flow rotates about the centre rather than how energetic the flow is. In the finite-volume implementation, the integrals in (2.18)–(2.19) are evaluated with the same control-volume weights used in the discrete kinetic-energy inner product. This diagnostic is used later to define the switching rule for the Rayleigh–Bénard ROMs.

2.2 Boussinesq approximation

Rayleigh–Bénard convection is driven by temperature-dependent density differences. In a fully general model, density variations would affect the mass, momentum and energy equations. However, in many thermal convection problems, the relative density variations are small, even though their effect on buoyancy is important. The Oberbeck–Boussinesq approximation uses this separation of effects [14, 37].

The approximation treats the material properties as constant and retains density variations only in the buoyancy force. The density is approximated by the linear relation

$$\rho(T) = \rho_0\,(1 - \alpha(T - T_0))\,, \tag{2.20}$$

where ρ_0 and T_0 denote reference density and temperature values, respectively. In the governing equations, the density is then replaced by ρ_0 everywhere except in the gravitational body force. After absorbing the hydrostatic contribution into a modified pressure, this produces the buoyancy term $g\,\alpha(T - T_0)\hat{\boldsymbol{e}}_y$ in the momentum equation, while the temperature field is transported by an advection–diffusion equation. The resulting continuous Rayleigh–Bénard model used by the full-order simulations is written explicitly in Section 3.1.

The approximation is valid when the relative density variation is small. This is commonly expressed as

$$\alpha \Delta T \ll 1. \tag{2.21}$$

Although density variations are assumed small, they are not ignored: they remain in the buoyancy term, which is precisely the term that drives thermal convection. This makes the Boussinesq approximation suitable for Rayleigh–Bénard convection, where the temperature field drives the velocity field through buoyancy, and the velocity field transports temperature through advection [4, 37].

2.3 Finite volume method

The finite volume method is a spatial discretization method based on the integral form of conservation laws. Instead of enforcing the governing equations pointwise, the computational domain is divided into a finite number of control volumes, and the balance laws are imposed on each control volume. This makes the method particularly natural for fluid dynamics, where mass, momentum and heat are transported through fluxes [15].

Let $q(\boldsymbol{x}, t)$ be a scalar conserved quantity, such as temperature, mass, or a component of momentum. Let $\boldsymbol{F}(q)$ denote its flux and $S(q)$ its source term. A generic local conservation law is

$$\frac{\partial q}{\partial t} + \boldsymbol{\nabla} \cdot \boldsymbol{F}(q) = S(q), \tag{2.22}$$

where $\boldsymbol{x}$ denotes spatial position. Integrating over a control volume $\Omega_i \subset \Omega$ gives

$$\frac{d}{dt} \int_{\Omega_i} q \, d\Omega + \int_{\Omega_i} \boldsymbol{\nabla} \cdot \boldsymbol{F}(q) \, d\Omega = \int_{\Omega_i} S(q) \, d\Omega. \tag{2.23}$$

Using the divergence theorem, the volume integral of the divergence becomes a surface integral over the cell boundary:

$$\int_{\Omega_i} \boldsymbol{\nabla} \cdot \boldsymbol{F}(q) \, d\Omega = \int_{\partial \Omega_i} \boldsymbol{F}(q) \cdot \boldsymbol{n} \, dS. \tag{2.24}$$

Here $\boldsymbol{n}$ is the outward unit normal on the control-volume boundary $\partial \Omega_i$, and dS denotes the corresponding boundary measure. Thus, the evolution of the cell average is determined by the net flux through the cell faces and by the sources inside the cell.

A key advantage of the finite volume method is local conservation. The flux leaving one cell through a shared face enters the neighbouring cell with the opposite sign. Therefore, if the face fluxes are treated consistently, the method conserves the discrete quantity locally as well as globally.

For incompressible flows, finite volume methods are often combined with staggered grids. In a staggered grid, pressure unknowns are stored at cell centres, while velocity components are stored at cell faces. This arrangement naturally connects the velocity fluxes through the cell faces with the incompressibility constraint

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0. \tag{2.25}$$

It also helps avoid pressure–velocity decoupling issues that can occur on collocated grids [1].

The numerical experiments in this thesis use an energy-conserving staggered finite-volume solver for the incompressible Navier–Stokes equations, following the discretization framework of Agdestein and Sanderse [1].

In this thesis, the finite volume discretization is important not only for the full-order simulation, but also for the construction of reduced-order models. If the full-order discretization

preserves important physical properties, such as mass conservation and the correct kinetic-energy behaviour, then these properties can be inherited more naturally by the reduced model. This is one motivation for using structure-preserving or energy-conserving finite volume discretizations in reduced-order modelling of incompressible flows [32].

2.4 Vorticity, kinetic energy and enstrophy

Several diagnostic quantities are used to interpret and validate the dipole–wall collision. The scalar vorticity in two dimensions is the out-of-plane component of the curl of the velocity field. It is defined as

$$\omega = \frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y}, \tag{2.26}$$

where $\boldsymbol{u} = (u_x, u_y)$. Vorticity measures the local rotation of the flow. In the dipole–wall collision, coherent vortices correspond to concentrated regions of positive or negative vorticity.

The total kinetic energy is defined by

$$E(t) = \frac{1}{2} \int_{\Omega} |\boldsymbol{u}(\boldsymbol{x}, t)|^2 \, d\Omega. \tag{2.27}$$

For an unforced viscous incompressible flow, kinetic energy is dissipated by viscosity. Therefore, the evolution of $E(t)$ is a useful global quantity for checking whether the simulation behaves physically.

The total enstrophy is defined by

$$\mathscr{Z}(t) = \frac{1}{2} \int_{\Omega} \omega^2(\boldsymbol{x}, t) \, d\Omega. \tag{2.28}$$

Enstrophy measures the intensity of vorticity in the flow. It is particularly useful in the dipole–wall collision because the no-slip wall generates thin layers of high vorticity. In the benchmark simulations of Clercx and Bruneau, peaks in the enstrophy correspond to the main vortex–wall collision events [13].

Kinetic energy and enstrophy are global quantities: they summarize the strength of the flow and vorticity, but not the location of the vortices or wall-generated vorticity layers. For this reason, validation of the dipole–wall collision should combine global diagnostics such as kinetic energy and enstrophy with visual inspection of the vorticity field.

Chapter 3

Full-Order Model and Problem Setup

Chapter 2 introduced the physical parameters and diagnostics used in the two test cases. This chapter now specifies the full-order models that generate the data for the reduced-order models. The purpose is to identify the continuous equations, the spatial discretization, and the polynomial structure that will later be used by projection and operator inference.

3.1 Continuous governing equations

The starting point is the incompressible Navier–Stokes equations. They express conservation of mass and momentum:

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0, \tag{3.1}$$

$$\underbrace{\frac{\partial \boldsymbol{u}}{\partial t}}_{\text{unsteady term}} + \underbrace{\boldsymbol{\nabla} \cdot (\boldsymbol{u} \otimes \boldsymbol{u})}_{\text{convective term}} = \underbrace{-\boldsymbol{\nabla} p}_{\text{pressure term}} + \underbrace{\nu \boldsymbol{\nabla} \cdot \left(\boldsymbol{\nabla} \boldsymbol{u} + (\boldsymbol{\nabla} \boldsymbol{u})^T \right)}_{\text{viscous term}} + \underbrace{\boldsymbol{f}}_{\text{forcing term}} . \tag{3.2}$$

Here, $\boldsymbol{u}(\xi, t)$ is the velocity field, $p(\xi, t)$ is the modified pressure divided by density, $\xi \in \Omega \subset \mathbb{R}^2$ denotes the spatial coordinate, t denotes time, $\nu \geq 0$ is the kinematic viscosity, and $\boldsymbol{f}$ represents body forces. The equations are supplemented with initial and boundary conditions [1].

For the dipole–wall collision test case, the forcing term is absent and the flow evolves from a prescribed vortex-dipole initial condition. Its main nondimensional control parameter is the Reynolds number $Re = UL/\nu$, introduced in Section 2.1. In the ROM state, only the velocity degrees of freedom are retained; pressure is used to enforce incompressibility rather than treated as an independent dynamical state.

For the Rayleigh–Bénard test case, the momentum equation is coupled to temperature through the Boussinesq approximation from Section 2.2. Using the same notation as in that section, the continuous model is

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0, \tag{3.3}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{\nabla} \cdot (\boldsymbol{u} \otimes \boldsymbol{u}) = -\boldsymbol{\nabla} p + \nu \boldsymbol{\nabla} \cdot \left(\boldsymbol{\nabla} \boldsymbol{u} + (\boldsymbol{\nabla} \boldsymbol{u})^T \right) + g\alpha(T - T_0)\widehat{\boldsymbol{e}}_y, \tag{3.4}$$

$$\frac{\partial T}{\partial t} + \boldsymbol{\nabla} \cdot (\boldsymbol{u} T) = \kappa \boldsymbol{\nabla}^2 T. \tag{3.5}$$

Here $\widehat{\boldsymbol{e}}_y$ denotes the vertical unit vector, while the material parameters and temperature notation are those introduced in Chapter 2. Thus, in contrast to the dipole–wall collision, the Rayleigh–Bénard state contains both velocity and temperature unknowns [2, 11].

3.2 Spatial discretization

Although the governing equations are formulated in continuous space and time, numerical simulations require a discrete representation. As discussed in Section 2.3, this thesis uses a finite-volume viewpoint: the domain is split into control volumes and the equations are advanced by balancing fluxes through cell faces. This choice is important for this work because the full-order solver uses a staggered, energy-conserving finite-volume discretization for incompressible flow [1, 32]. This matters for the ROM construction because POD modes and Galerkin projections depend on the inner product used to measure state size and orthogonality. The same discrete inner products and control-volume weights are therefore used when constructing POD bases and projecting full-order quantities, so that reduced quantities are consistent with the energy and conservation structure of the full-order discretization.

3.3 Polynomial representation

After spatial discretization, the governing equations can be interpreted as a finite-dimensional dynamical system for the discrete state

$$\boldsymbol{x}(t) \in \mathbb{R}^N,$$

where N denotes the number of state degrees of freedom. For the dipole–wall collision problem, $\boldsymbol{x}$ consists of the velocity degrees of freedom. For the Rayleigh–Bénard problem, $\boldsymbol{x}$ consists of the coupled velocity-temperature state,

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{T} \end{bmatrix}.$$

The pressure is not included as an additional nonlinear state variable. Instead, following the discretize-first viewpoint used in the solver, incompressibility is enforced at the discrete level by the pressure-projection treatment of the velocity field [1, 32]. This is the same full-order structure that the intrusive and operator-inference ROMs sample later.

For constant viscosity and thermal diffusivity, diffusion is linear in the state. The velocity convection term contains products of velocity components, and in Rayleigh–Bénard convection the temperature equation adds quadratic thermal-advection terms. Consequently, the semi-discrete systems considered in this thesis have polynomial structure of degree two in the discrete state. They can be written abstractly in affine-quadratic form as

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{c} + A_1 \boldsymbol{x}(t) + A_2 \boldsymbol{x}^2(t), \tag{3.6}$$

where

$$\boldsymbol{c} \in \mathbb{R}^N, \qquad A_1 \in \mathbb{R}^{N \times N}, \qquad A_2 \in \mathbb{R}^{N \times N_2}.$$

Here, A_1 represents the linear contribution, associated for instance with diffusion, buoyancy coupling, and the incompressibility treatment, while A_2 represents the quadratic convective and advective interactions. The vector $\boldsymbol{c}$ represents affine contributions independent of the solution, for example those induced by non-homogeneous boundary conditions.

For the dipole–wall collision case, this affine term is absent and one may take $\boldsymbol{c} = \boldsymbol{0}$. For the Rayleigh–Bénard case, the non-homogeneous temperature boundary conditions can lead to a non-zero affine contribution when the absolute temperature field is used as part of the state. Alternatively, one could introduce a temperature lifting function satisfying the non-homogeneous thermal boundary conditions and formulate the ROM in terms of the homogeneous temperature fluctuation, thereby absorbing the corresponding affine contribution into the lifted equations. This approach was not used in this project to keep the

FOM and ROM state variables identical and to avoid introducing additional lift-dependent operators in the reduced model [28, 32].

The vector

$$\boldsymbol{x}^2(t) \in \mathbb{R}^{N_2}, \qquad N_2 = \binom{N+1}{2},$$

contains the unique quadratic products of the entries of $\boldsymbol{x}(t)$, constructed from the Kronecker product $\boldsymbol{x}(t) \otimes \boldsymbol{x}(t)$ after removing duplicate entries due to commutativity of multiplication [26, 29].

Chapter 4

Reduced-Order Model

The full-order model introduced in Chapter 3 provides a high-dimensional approximation of the governing incompressible flow equations. Although this representation is suitable for accurate numerical simulation, the number of degrees of freedom N can be very large. Reduced-order modelling replaces this high-dimensional system by a lower-dimensional dynamical system that preserves the dominant behaviour of the original flow while being significantly cheaper to evaluate.

The first step is to choose the low-dimensional space in which the reduced state will evolve. Rather than selecting this space a priori, this thesis uses data from full-order simulations to identify dominant flow structures. Proper orthogonal decomposition (POD) provides such a basis by finding modes that optimally represent the available snapshot data in an energy-weighted norm. Once this basis is available, the full state can be approximated by a small number of time-dependent coefficients.

The second step is to determine how these reduced coefficients evolve in time. Operator inference addresses this task by learning reduced operators from data while retaining a polynomial structure motivated by the governing equations [27, 23]. Exact operator inference is important in this thesis because it provides a controlled way to compare data-driven operators with intrusive projected operators: under suitable rank and consistency assumptions, the inferred operators recover the intrusive reduced operators up to numerical precision [29].

Most of the chapter first describes ROM construction for one fixed reduced basis. The final part then considers a regime-local extension for flows whose dominant dynamics change in time. Such changes can occur when coherent structures translate, rotate, merge, or when the flow moves from one dominant pattern to another. Several related approaches address this limitation in different ways. Peherstorfer [25] adapts the reduced basis online in time for transport-dominated problems by updating local reduced spaces from selected full-model samples. Grimberg et al. [17] use local reduced-order bases together with hyperreduction for nonlinear Petrov–Galerkin ROMs, so that different parts of a convection-dominated solution manifold can be represented by different affine subspaces. Anderson et al. [3] introduce space-local bases, in which the computational domain is partitioned and sparse local basis functions are used to improve efficiency for large turbulent-flow ROMs. The approach in this thesis is different: the square Rayleigh–Bénard case is not split spatially, and the basis is not adapted online. Instead, the dominant change is a time-local change in the orientation of the large-scale circulation, so several local ROMs are trained on preselected time regimes and switched according to fixed times obtained from the normalized angular-momentum history introduced in Section 2.1.5. This keeps the method simple and reproducible while matching the main regime structure observed in the reference trajectory.

4.1 Proper Orthogonal Decomposition

Proper orthogonal decomposition is used here as the bridge between full-order simulation data and a reduced state description. In fluid mechanics, POD has been widely used to identify energetic coherent structures and to build low-dimensional flow models [5, 36]. For example, POD modes can represent dominant vortical structures in wake flows, energetic structures in turbulent data, or thermal structures in convection problems. In this thesis, POD is not the final ROM by itself; it provides the reduced coordinates on which the intrusive, operator-inference and switched ROMs are built.

By evolving the full-order model over a prescribed simulation time interval $[0, T]$, one obtains a collection of solution snapshots at discrete time instances

$$0 = t_1 < t_2 < \cdots < t_s = T.$$

Each snapshot corresponds to a full-order state

$$\boldsymbol{x}(t_k) \in \mathbb{R}^N,$$

where N denotes the number of spatial degrees of freedom.
These snapshots are collected column-wise in the snapshot matrix

$$\mathbf{X} = \begin{bmatrix} \boldsymbol{x}(t_1) & \boldsymbol{x}(t_2) & \cdots & \boldsymbol{x}(t_s) \end{bmatrix} \in \mathbb{R}^{N \times s}. \tag{4.1}$$

Since the unknowns are defined on a staggered (and possibly stretched) grid, the Euclidean inner product on $\mathbb{R}^N$ is not necessarily consistent with the corresponding discrete energy inner product. Therefore, a weighted inner product is introduced by means of a diagonal weight matrix $\mathbf{M} \in \mathbb{R}^{N \times N}$,

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\mathbf{M}} := \boldsymbol{x}^T \mathbf{M} \boldsymbol{y}, \qquad \|\boldsymbol{x}\|_{\mathbf{M}}^2 = \boldsymbol{x}^T \mathbf{M} \boldsymbol{x}. \tag{4.2}$$

For a velocity-only state, the diagonal entries of $\mathbf{M}$ represent the appropriate control-volume sizes associated with each staggered velocity degree of freedom [32]. With this choice, $\|\boldsymbol{x}\|_{\mathbf{M}}^2$ is proportional to the discrete kinetic energy. For the Rayleigh–Bénard case, the state contains both velocity and temperature unknowns, so $\mathbf{M}$ is understood as a state-weight matrix containing the corresponding velocity and scalar control-volume weights.

The goal is to identify a low-dimensional subspace that captures the most energetic structures present in the snapshot data (measured in the $\mathbf{M}$-inner product). This is achieved by applying a singular value decomposition (SVD) to the weighted snapshot matrix,

$$\mathbf{M}^{1/2} \mathbf{X} = \mathbf{U} \boldsymbol{\Sigma} \mathbf{W}^T, \tag{4.3}$$

where, for the compact SVD,

$$\mathbf{U} \in \mathbb{R}^{N \times m}, \qquad \boldsymbol{\Sigma} \in \mathbb{R}^{m \times m}, \qquad \mathbf{W} \in \mathbb{R}^{s \times m}, \qquad m = \min(N, s).$$

The columns of $\mathbf{U}$ are the left singular vectors, the diagonal entries of $\boldsymbol{\Sigma}$ are the singular values, and the columns of $\mathbf{W}$ are the right singular vectors [19]. The dominant reduced basis in the original coordinates is obtained by retaining the first r weighted left singular vectors, with $r \ll N$:

$$\boldsymbol{\Phi} = \mathbf{M}^{-1/2} \begin{bmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_r \end{bmatrix} \in \mathbb{R}^{N \times r}. \tag{4.4}$$

The columns of $\boldsymbol{\Phi}$ are referred to as the POD modes and form an orthonormal basis of the reduced space with respect to the $\mathbf{M}$-inner product,

$$\boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{\Phi} = \mathbf{I}_r. \tag{4.5}$$

The POD basis is optimal, in the sense that it minimizes the projection error of the snapshot data (measured in the $\mathbf{M}$-norm) among all r-dimensional $\mathbf{M}$-orthonormal bases [40, 8]:

$$\mathbf{\Phi} = \arg\min_{\mathbf{V}\in\mathbb{R}^{N\times r}} \left\|\mathbf{M}^{1/2}\left(\mathbf{X}-\mathbf{V}\mathbf{V}^T\mathbf{M}\mathbf{X}\right)\right\|_F^2, \qquad \text{subject to } \mathbf{V}^T\mathbf{M}\mathbf{V}=\mathbf{I}_r. \tag{4.6}$$

Using the POD basis, the full-order state is approximated by a mean state plus a linear combination of r reduced coordinates:

$$\boldsymbol{x}(t) \approx \boldsymbol{x}_r(t) := \boldsymbol{\mu} + \mathbf{\Phi}\boldsymbol{a}(t), \tag{4.7}$$

where $\boldsymbol{\mu}=\mathbf{0}$ for an uncentered POD basis and $\boldsymbol{\mu}=\overline{\boldsymbol{x}}$ for a centered POD basis. The vector

$$\boldsymbol{a}(t) = \begin{bmatrix} a_1(t) & a_2(t) & \cdots & a_r(t)\end{bmatrix}^T \in \mathbb{R}^r$$

is the reduced state vector. Since the basis is $\mathbf{M}$-orthonormal, the reduced coordinates are obtained by $\mathbf{M}$-orthogonal projection of the deviation from the mean:

$$\boldsymbol{a}(t) = \mathbf{\Phi}^T\mathbf{M}\left(\boldsymbol{x}(t)-\boldsymbol{\mu}\right). \tag{4.8}$$

Equations (4.7) and (4.8) define the algebraic connection between the full-order and reduced-order states.

4.1.1 Choice of the ROM Dimension

In practice, the dimension r can be selected by prescribing a retained energy threshold,

$$\frac{\sum_{i=1}^{r}\sigma_i^2}{\sum_{i=1}^{m}\sigma_i^2} \geq \eta, \tag{4.9}$$

where σ_i are the singular values and η is typically chosen close to one, for example $\eta = 0.99$ [3].

For flows with a large steady or slowly varying mean component, such as the Rayleigh–Bénard problem considered here, the retained energy criterion must be interpreted with care. The snapshot matrix $\mathbf{X}$ contains the full state, including the mean temperature distribution imposed by the thermal boundary conditions. Consequently, the first POD modes may capture a large fraction of the total $\mathbf{M}$-weighted energy simply by representing this mean or background component, while still failing to resolve the dynamically important convective fluctuations.

To assess the representation of these fluctuations, one may introduce the temporal mean state

$$\overline{\boldsymbol{x}} = \frac{1}{s}\sum_{k=1}^{s}\boldsymbol{x}(t_k), \tag{4.10}$$

and the corresponding fluctuation snapshot matrix

$$\mathbf{X}' = \begin{bmatrix}\boldsymbol{x}(t_1)-\overline{\boldsymbol{x}} & \boldsymbol{x}(t_2)-\overline{\boldsymbol{x}} & \cdots & \boldsymbol{x}(t_s)-\overline{\boldsymbol{x}}\end{bmatrix}. \tag{4.11}$$

In the following, a centered POD basis refers to a basis constructed from $\mathbf{X}'$, so that the reduced coordinates represent deviations from $\overline{\boldsymbol{x}}$ and the mean is added back when reconstructing full states, as in (4.7). The retained fluctuation energy is then defined analogously as

$$\frac{\sum_{i=1}^{r}(\sigma_i')^2}{\sum_{i=1}^{m}(\sigma_i')^2},$$

where σ_i' are the singular values of $\mathbf{M}^{1/2}\mathbf{X}'$. This quantity measures how well the POD basis captures deviations from the mean state rather than the total raw state energy.

This distinction is important in the Rayleigh–Bénard simulations. A threshold such as $\eta = 0.99$ applied to the raw snapshots can lead to a very small basis, because the dominant mean thermal structure contains most of the energy. Such a basis may have a small projection error in the raw energy norm but still give a poor visual or dynamical representation of plumes, boundary layers, and Nusselt number evolution. Therefore, in addition to the raw retained energy, the retained fluctuation energy and validation quantities such as kinetic energy, Nusselt numbers, and reconstruction error should be inspected when selecting the ROM dimension.

Once the reduced basis and dimension have been selected, the remaining task is to determine how the reduced coordinates evolve in time. The next section derives the polynomial reduced model obtained when the full-order equations have affine, linear, and quadratic contributions.

4.2 Reduced Polynomial Dynamical System

The POD approximation can now be combined with the polynomial full-order model introduced in Equation (3.6). To avoid confusion between full-order and reduced affine terms, we denote the full-order affine contribution in this paragraph by $\boldsymbol{c}_{\mathrm{FOM}}$. Thus, at the semi-discrete level one may write the full-order dynamics abstractly as

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{c}_{\mathrm{FOM}} + A_1 \boldsymbol{x}(t) + A_2 \boldsymbol{x}^2(t),$$

where $A_1 \in \mathbb{R}^{N\times N}$, $A_2 \in \mathbb{R}^{N\times N_2}$, and $N_2 = \binom{N+1}{2}$. Using the POD approximation

$$\boldsymbol{x}(t) \approx \boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}(t), \qquad \dot{\boldsymbol{x}}(t) \approx \boldsymbol{\Phi}\dot{\boldsymbol{a}}(t),$$

gives, formally,

$$\boldsymbol{\Phi}\dot{\boldsymbol{a}}(t) \approx \boldsymbol{c}_{\mathrm{FOM}} + A_1\left(\boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}(t)\right) + A_2\left(\boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}(t)\right)^2.$$

The Galerkin projection requires the residual of this approximation to be orthogonal to the reduced space with respect to the $\mathbf{M}$-inner product. Therefore, multiplying by $\boldsymbol{\Phi}^T\mathbf{M}$ and using the $\mathbf{M}$-orthonormality relation $\boldsymbol{\Phi}^T\mathbf{M}\boldsymbol{\Phi} = I_r$, one obtains the formal projected system

$$\dot{\boldsymbol{a}}(t) = \boldsymbol{\Phi}^T\mathbf{M}\boldsymbol{c}_{\mathrm{FOM}} + \boldsymbol{\Phi}^T\mathbf{M}A_1\left(\boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}(t)\right) + \boldsymbol{\Phi}^T\mathbf{M}A_2\left(\boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}(t)\right)^2.$$

For an uncentered basis, $\boldsymbol{\mu} = \mathbf{0}$, and the projected operators can be written directly in terms of the full-order polynomial matrices. The quadratic term $(\boldsymbol{\Phi}\boldsymbol{a}(t))^2$ contains the unique quadratic products of the entries of the lifted state $\boldsymbol{\Phi}\boldsymbol{a}(t)$. Since these products are quadratic functions of the reduced coordinates, there exists a matrix

$$P_{\boldsymbol{\Phi}} \in \mathbb{R}^{N_2\times r_2}, \qquad r_2 = \binom{r+1}{2},$$

such that

$$(\boldsymbol{\Phi}\boldsymbol{a}(t))^2 = P_{\boldsymbol{\Phi}}\boldsymbol{a}^2(t),$$

where $\boldsymbol{a}^2(t) \in \mathbb{R}^{r_2}$ contains the unique quadratic products of the reduced coordinates. If the full-order polynomial operators were explicitly assembled, the corresponding reduced operators would therefore be

$$\widetilde{\boldsymbol{c}} = \boldsymbol{\Phi}^T\mathbf{M}\boldsymbol{c}_{\mathrm{FOM}}, \qquad \widetilde{A}_1 = \boldsymbol{\Phi}^T\mathbf{M}A_1\boldsymbol{\Phi}, \qquad \widetilde{A}_2 = \boldsymbol{\Phi}^T\mathbf{M}A_2P_{\boldsymbol{\Phi}}.$$

For a centered basis, the same affine-quadratic reduced structure is retained; the mean $\boldsymbol{\mu}$ shifts part of the full-order linear and quadratic contributions into the reduced affine and linear terms.

In the implementation, the full-order matrices A_1, A_2, and $\boldsymbol{c}_{\mathrm{FOM}}$ are not explicitly stored. This is especially important for the quadratic convective operator, whose full-order tensor

would be prohibitively large. Explicit construction of such operators is feasible in smaller one-dimensional settings [29], but not in the two-dimensional flow problems considered here. Instead, the reduced operators are obtained from projected evaluations of the implemented full-order routines. Reduced states are lifted to full-order variables, the appropriate FOM right-hand side or one-step map is evaluated, and the result is immediately projected with $\mathbf{\Phi}^T\mathbf{M}$. This query-and-project procedure is used for both flow problems: in the dipole–wall case with the velocity-only incompressible Navier–Stokes operator, and in the Rayleigh–Bénard case with the coupled Boussinesq operator and the same boundary treatment and pressure projection as in the FOM. In both cases the reduced model has the form

$$\dot{\boldsymbol{a}}(t) = \widetilde{\boldsymbol{c}} + \widetilde{A}_1 \boldsymbol{a}(t) + \widetilde{A}_2 \boldsymbol{a}^2(t), \tag{4.12}$$

where

$$\widetilde{\boldsymbol{c}} \in \mathbb{R}^r, \qquad \widetilde{A}_1 \in \mathbb{R}^{r\times r}, \qquad \widetilde{A}_2 \in \mathbb{R}^{r\times r_2}, \qquad r_2 = \binom{r+1}{2}.$$

The affine term is absent for the dipole–wall collision case, whereas for the Rayleigh–Bénard case it may be nonzero due to the non-homogeneous temperature boundary conditions when the absolute temperature field is included in the state.

For a fixed full-order problem, this is the point where intrusive projection and operator inference separate. In an intrusive ROM, the operators in (4.12) are computed from internal FOM right-hand-side and projection routines. The Rayleigh–Bénard intrusive reference should therefore be interpreted as a sampled Galerkin projection of the implemented semi-discrete FOM right-hand side, not as an explicit assembly of continuous weak-form matrices or of the full finite-volume quadratic tensor. The next section discusses how operators of the same form can instead be identified from projected data or time-stepper queries.

4.3 Exact Operator Inference

Operator inference determines the reduced operators in (4.12) directly from data, rather than by explicitly assembling and projecting the full-order polynomial operators. In its standard form, one collects a reduced trajectory $\{\boldsymbol{a}(t_k)\}_{k=1}^s$ together with approximations of the reduced time derivatives $\{\dot{\boldsymbol{a}}(t_k)\}_{k=1}^s$ and solves a regression problem for the unknown reduced operators. A practical difficulty is that the resulting regression matrix can be rank-deficient or ill-conditioned, so regularization is often introduced and its strength must be tuned [24]. This can lead to representations of the flow that differ considerably from the ones that would be obtained with the projected operators [34, 29].

Exact operator inference addresses this issue by choosing the training data such that the regression problem is square and well-posed, leading to a unique unregularized solution. Up to machine precision, and when the reduced model structure matches the true projected dynamics, this procedure recovers the same reduced operators that would be obtained by an intrusive projection [29]. In the implementation considered here, exact operator inference still requires queries to the full-order time-stepper, but it avoids explicitly assembling or projecting the full-order polynomial matrices.

Regression formulation

Evaluating the affine-quadratic reduced model (4.12) at discrete times t_k gives

$$\dot{\boldsymbol{a}}(t_k) = \boldsymbol{c} + \widetilde{A}_1 \boldsymbol{a}(t_k) + \widetilde{A}_2 \boldsymbol{a}^2(t_k), \qquad k = 1, \ldots, s, \tag{4.13}$$

where $\boldsymbol{a}^2(t_k) \in \mathbb{R}^{r_2}$ contains the unique quadratic products of the reduced coordinates and $r_2 = \binom{r+1}{2}$. If no affine term is included, then $\boldsymbol{c} = \boldsymbol{0}$ and the constant feature is omitted from the regression.

Define the data matrices

$$\mathbf{A} = \begin{bmatrix} \boldsymbol{a}(t_1) & \cdots & \boldsymbol{a}(t_s) \end{bmatrix} \in \mathbb{R}^{r \times s}, \qquad \dot{\mathbf{A}} = \begin{bmatrix} \dot{\boldsymbol{a}}(t_1) & \cdots & \dot{\boldsymbol{a}}(t_s) \end{bmatrix} \in \mathbb{R}^{r \times s}, \tag{4.14}$$

and similarly

$$\mathbf{A}^{(2)} = \begin{bmatrix} \boldsymbol{a}^2(t_1) & \cdots & \boldsymbol{a}^2(t_s) \end{bmatrix} \in \mathbb{R}^{r_2 \times s}.$$

For the purely quadratic case, the feature matrix and operator matrix are

$$\mathbf{Z} = \begin{bmatrix} \mathbf{A} \\ \mathbf{A}^{(2)} \end{bmatrix} \in \mathbb{R}^{(r+r_2) \times s}, \qquad \widetilde{\mathbf{O}} = \begin{bmatrix} \widetilde{A}_1 & \widetilde{A}_2 \end{bmatrix} \in \mathbb{R}^{r \times (r+r_2)}. \tag{4.15}$$

For the affine-quadratic case, the constant feature is added:

$$\mathbf{Z} = \begin{bmatrix} \mathbf{1} \\ \mathbf{A} \\ \mathbf{A}^{(2)} \end{bmatrix} \in \mathbb{R}^{(1+r+r_2) \times s}, \qquad \widetilde{\mathbf{O}} = \begin{bmatrix} \boldsymbol{c} & \widetilde{A}_1 & \widetilde{A}_2 \end{bmatrix} \in \mathbb{R}^{r \times (1+r+r_2)}. \tag{4.16}$$

In both cases, the snapshot relations can be written compactly as

$$\dot{\mathbf{A}} = \widetilde{\mathbf{O}}\,\mathbf{Z}. \tag{4.17}$$

In standard operator inference, one typically computes or approximates $\dot{\boldsymbol{a}}(t_k)$ from available data and determines $\widetilde{\mathbf{O}}$ by least squares, optionally with Tikhonov regularization:

$$\widetilde{\mathbf{O}} = \arg\min_{\mathbf{Y}} \left\| \dot{\mathbf{A}} - \mathbf{Y}\mathbf{Z} \right\|_F^2 + \lambda \left\| \mathbf{Y} \right\|_F^2. \tag{4.18}$$

In the present implementation, trajectory-based operator inference uses projected full-order derivative samples when they are available, rather than relying only on finite differences of the reduced trajectory.

A practical limitation of this trajectory-based formulation is that the number of independent trajectory samples bounds the identifiable reduced dimension. For a quadratic ROM without a constant term, the feature matrix contains

$$n_f = r + \frac{r(r+1)}{2}$$

rows; for an affine-quadratic ROM, one additional constant row is present. If only s trajectory samples are available and $s < n_f$, the least-squares problem is underdetermined. Tikhonov regularization can select one stable solution of this underdetermined problem, but it cannot add the missing independent information needed to identify all quadratic coefficients. Therefore, when the requested POD energy threshold implies a rank for which $n_f > s$, a conventional trajectory-based OpInf fit must either reduce the rank or accept an operator fit that is mostly set by regularization. This distinction is important in the numerical results: a lower fitted rank can make the online ROM cheaper, but it also means that the fitted model has not actually retained the requested POD energy.

Exact operator inference

Exact operator inference sets $\lambda = 0$ and instead constructs training data such that $\mathbf{Z}$ has full row rank. Denoting the number of features by

$$n_f = \begin{cases} r + r_2, & \text{quadratic case,} \\ 1 + r + r_2, & \text{affine-quadratic case,} \end{cases} \tag{4.19}$$

a particularly simple situation is to choose $s = n_f$ samples such that $\mathbf{Z} \in \mathbb{R}^{n_f \times n_f}$ is nonsingular. Then (4.17) becomes a square linear system with the unique solution

$$\widetilde{\mathbf{O}} = \dot{\mathbf{A}}\mathbf{Z}^{-1}. \tag{4.20}$$

The reduced operators are obtained by splitting $\widetilde{\mathbf{O}}$ according to either (4.15) or (4.16).

The remaining question is how to generate a set of samples that makes $\mathbf{Z}$ nonsingular. The key idea is to avoid relying on a single long trajectory and instead generate dedicated snapshot pairs from multiple one-step simulations:

1. Choose reduced training states $\{\boldsymbol{a}_0^{(j)}\}_{j=1}^{n_f}$ such that the associated feature vectors span the feature space. For the quadratic case,

$$\boldsymbol{z}^{(j)} = \begin{bmatrix} \boldsymbol{a}_0^{(j)} \\ (\boldsymbol{a}_0^{(j)})^2 \end{bmatrix} \in \mathbb{R}^{r+r_2},$$

   whereas for the affine-quadratic case,

$$\boldsymbol{z}^{(j)} = \begin{bmatrix} 1 \\ \boldsymbol{a}_0^{(j)} \\ (\boldsymbol{a}_0^{(j)})^2 \end{bmatrix} \in \mathbb{R}^{1+r+r_2}.$$

   In the implementation, the rank-ensuring states are chosen deterministically. For the quadratic case they consist of the coordinate vectors $\boldsymbol{e}_j$, the sums $\boldsymbol{e}_j + \boldsymbol{e}_k$ for $j < k$, and the diagonal samples $2\boldsymbol{e}_j$. In the affine-quadratic case, the zero reduced state is added as the first sample. This choice is not meant to isolate each coefficient individually, but to make the feature matrix nonsingular [29].

2. Lift each reduced state to the full space,

$$\boldsymbol{x}_0^{(j)} = \boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}_0^{(j)} \in \mathbb{R}^N, \tag{4.21}$$

   where $\boldsymbol{\mu} = \mathbf{0}$ for an uncentered basis and $\boldsymbol{\mu} = \overline{\boldsymbol{x}}$ from (4.10) when a centered basis, constructed from the fluctuation snapshots in (4.11), is used. The same essential boundary treatment used by the full-order model is then applied when required, the full-order model is advanced by a single time step Δt, and the resulting state is denoted by $\boldsymbol{x}_1^{(j)}$. In the implementation, the one-step size Δt used for these training queries is chosen from the observed time scale of the projected reduced trajectory, so that the one-step perturbation is not excessively large. This estimate is also capped by the FOM snapshot spacing.

3. Form reduced time-derivative data using the same one-step discretization. With an explicit-Euler-consistent difference quotient this reads

$$\dot{\boldsymbol{a}}^{(j)} \approx \boldsymbol{\Phi}^T \mathbf{M} \frac{\boldsymbol{x}_1^{(j)} - \boldsymbol{x}_0^{(j)}}{\Delta t}. \tag{4.22}$$

   Exact recovery requires these one-step training data to be generated with an explicit-Euler-consistent update; this is the setting assumed in the exact operator inference analysis [29].

Stacking $\{\boldsymbol{a}_0^{(j)}\}$ and $\{\dot{\boldsymbol{a}}^{(j)}\}$ into matrices of the form (4.14) produces $\mathbf{Z}$ and $\dot{\mathbf{A}}$ for which (4.20) can be applied. In this way, the reduced operators are inferred by a direct linear solve, without regularization parameters and without requiring long time histories [29].

This construction moves a non-negligible part of the computational effort to the offline stage. Before any online ROM prediction can be made, the FOM trajectory must be generated and stored, a POD basis must be computed, and the exact-OpInf queries in the reduced

feature space must be evaluated. For an affine-quadratic ROM of dimension r, exact OpInf requires $n_f = 1+r+r(r+1)/2$ one-step FOM queries, followed by the solution of a dense $n_f \times n_f$ operator-identification problem; in the purely quadratic case the constant query is omitted. Thus the online speed-ups reported later should be interpreted as amortized gains: they are most meaningful when the same trained ROM is reused for many parameter studies, design evaluations, or long-time continuations, while the one-time FOM, POD and operator-fitting costs remain part of the total model-construction budget.

Exact reconstruction of the intrusive reduced operators

The main appeal of exact operator inference is that, under ideal conditions, it recovers the same reduced operators that one would obtain by an intrusive Galerkin projection. For problems with essential boundary conditions, the full-order right-hand side below should be understood as the discrete full-order right-hand side evaluated with the same boundary treatment and pressure projection as in the FOM. The following theorem formalizes the exact-recovery statement in the notation of this chapter.

Theorem 1. *Consider a full-order model of the form*

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}(t)), \tag{4.23}$$

and let $\boldsymbol{\Phi} \in \mathbb{R}^{N\times r}$ be a $\mathbf{M}$-orthonormal POD basis satisfying (4.5). Let $\boldsymbol{\mu} = \mathbf{0}$ for an uncentered basis, and let $\boldsymbol{\mu} = \overline{\boldsymbol{x}}$ from (4.10) when a centered basis is constructed from (4.11). Define the intrusive Galerkin-projected reduced dynamics by

$$\dot{\boldsymbol{a}}(t) = \boldsymbol{\Phi}^T \mathbf{M}\, \boldsymbol{f}(\boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}(t)). \tag{4.24}$$

Assume that the projected right-hand side admits the affine-quadratic polynomial representation

$$\boldsymbol{\Phi}^T \mathbf{M}\, \boldsymbol{f}(\boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}) = \boldsymbol{c} + \widetilde{A}_1 \boldsymbol{a} + \widetilde{A}_2 \boldsymbol{a}^2, \tag{4.25}$$

with $\boldsymbol{c} = \mathbf{0}$ in the purely quadratic case. Choose $s = n_f$ training states $\boldsymbol{a}_0^{(j)}$, $j = 1,\dots,n_f$, such that the associated feature matrix $\mathbf{Z}$ is nonsingular. For each j, lift $\boldsymbol{a}_0^{(j)}$ to $\boldsymbol{x}_0^{(j)} = \boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}_0^{(j)}$, after applying the same boundary treatment if required, and generate a single explicit-Euler full-order step with time step $\Delta t > 0$:

$$\boldsymbol{x}_1^{(j)} = \boldsymbol{x}_0^{(j)} + \Delta t\, \boldsymbol{f}(\boldsymbol{x}_0^{(j)}). \tag{4.26}$$

Define reduced derivative data by the Euler-consistent quotient

$$\dot{\boldsymbol{a}}^{(j)} = \boldsymbol{\Phi}^T \mathbf{M} \frac{\boldsymbol{x}_1^{(j)} - \boldsymbol{x}_0^{(j)}}{\Delta t}. \tag{4.27}$$

Then the unregularized operator inference problem (4.17) with $s = n_f$ has the unique solution $\widetilde{\mathbf{O}} = \dot{\mathbf{A}}\mathbf{Z}^{-1}$, and this solution coincides with the intrusive reduced operators in (4.25).

Proof. Fix $j \in \{1,\dots,n_f\}$. Using (4.26) in (4.27) yields

$$\dot{\boldsymbol{a}}^{(j)} = \boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{f}(\boldsymbol{x}_0^{(j)}) = \boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{f}(\boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}_0^{(j)}).$$

By the affine-quadratic structure assumption,

$$\dot{\boldsymbol{a}}^{(j)} = \boldsymbol{c} + \widetilde{A}_1 \boldsymbol{a}_0^{(j)} + \widetilde{A}_2 (\boldsymbol{a}_0^{(j)})^2.$$

Stacking these identities for $j = 1,\dots,n_f$ gives the matrix relation

$$\dot{\mathbf{A}} = \widetilde{\mathbf{O}}\mathbf{Z}.$$

Since $\mathbf{Z}$ is nonsingular by assumption, this linear system has the unique solution $\widetilde{\mathbf{O}} = \dot{\mathbf{A}}\mathbf{Z}^{-1}$, which is precisely the operator inference solution in the square, unregularized case. Therefore, the inferred operators coincide with the intrusive reduced operators. □

4.4 Angular-Momentum-Based Switched ROM

The exact operator-inference procedure described above constructs one reduced model for one chosen POD basis and one chosen training interval. This is sufficient when the trajectory remains in a single dynamical regime. For flows with time-local dynamics, however, a single global basis may have to represent structures that appear in different locations, orientations, or regimes. Examples include transport-dominated flows with moving coherent structures, wake flows with regime changes, and convection problems in which the dominant circulation changes direction [25, 17, 7]. A ROM trained only on one part of such a trajectory may be inaccurate once the flow leaves that regime, while a global ROM may require too many modes to represent all regimes accurately.

To address this issue in the Rayleigh–Bénard test case, some runs use a scheduled switched ROM. The idea is to train several local ROMs on different time windows of the same FOM trajectory, and then use a fixed schedule to decide which local ROM is active during prediction. In the implementation used here there are three local models, corresponding to the regimes

$$m \in \{\text{pre}, \text{transition}, \text{post}\}.$$

Each local model has its own POD basis, mean state and inferred operators,

$$\boldsymbol{\Phi}_m, \qquad \boldsymbol{\mu}_m, \qquad \boldsymbol{c}_m, \qquad \widetilde{A}_{1,m}, \qquad \widetilde{A}_{2,m}.$$

Thus, while model m is active, the reconstructed state and reduced dynamics are

$$\boldsymbol{x}_m(t) = \boldsymbol{\mu}_m + \boldsymbol{\Phi}_m \boldsymbol{a}_m(t), \qquad \dot{\boldsymbol{a}}_m(t) = \boldsymbol{c}_m + \widetilde{A}_{1,m}\boldsymbol{a}_m(t) + \widetilde{A}_{2,m}\boldsymbol{a}_m^2(t). \tag{4.28}$$

The local reduced dimensions are allowed to be different, so the coefficient vector $\boldsymbol{a}_m$ should be understood as coordinates in the currently active reduced space, not as one fixed coordinate vector shared by all regimes.

The schedule is derived from the normalized angular momentum ℓ_z defined in (2.19). This quantity is useful here because its sign identifies the dominant orientation of the large-scale circulation, while values close to zero indicate that the circulation has weakened or that the flow is passing through a reversal. The normalized angular momentum is evaluated on the reference FOM trajectory before the ROM prediction is performed. Let $t_{\text{pre}\to\text{tr}}$ be the first downward FOM crossing of the pre-to-transition threshold and let $t_{\text{tr}\to\text{post}}$ be the first downward FOM crossing of the transition-to-post threshold after that time. These two times define the switching schedule used by the ROM:

$$m(t) = \begin{cases} \text{pre}, & t < t_{\text{pre}\to\text{tr}}, \\ \text{transition}, & t_{\text{pre}\to\text{tr}} \le t < t_{\text{tr}\to\text{post}}, \\ \text{post}, & t \ge t_{\text{tr}\to\text{post}}. \end{cases}$$

During the online ROM simulation, the currently active local ROM is advanced over the next time interval and the active model is changed only when the fixed schedule time has been reached. Thus, the ROM does not use its own predicted angular momentum to decide when to switch. In practice the switch is applied at the first available ROM output time at or after the scheduled crossing time.

This scheduled choice differs from an online threshold rule in which the reconstructed ROM state would be used to compute ℓ_z and trigger model changes. Such a rule is sensitive to drift in the ROM angular-momentum trajectory and can generate switches at times that are inconsistent with the training windows. In the scheduled formulation used here, the angular-momentum thresholds are used only offline to locate the two FOM transition times; the online ROM then follows the resulting monotone sequence

$$\text{pre} \to \text{transition} \to \text{post}.$$

Let m_k denote the active regime after the k-th accepted output time and let

$$m_{k+1} = \begin{cases} \text{transition}, & m_k = \text{pre and } t_{k+1} \geq t_{\text{pre}\to\text{tr}}, \\ \text{post}, & m_k = \text{transition and } t_{k+1} \geq t_{\text{tr}\to\text{post}}, \\ m_k, & \text{otherwise.} \end{cases}$$

When the target regime changes from m to n, the coefficient vector is not transformed directly by a reduced matrix. Instead, the current state is first reconstructed in the full-order coordinates using the old basis, and that reconstructed state is then projected onto the new local basis:

$$\boldsymbol{x}_m = \boldsymbol{\mu}_m + \boldsymbol{\Phi}_m \boldsymbol{a}_m, \qquad \boldsymbol{a}_n = \boldsymbol{\Phi}_n^T \mathbf{M}\left(\boldsymbol{x}_m - \boldsymbol{\mu}_n\right). \tag{4.29}$$

This is the actual change of basis used by the switched ROM. It is a projection through the full state space, which is important because the local bases may have different dimensions and may have been constructed from different centered snapshot windows. After this reprojection, the simulation continues with the reduced operators of the new active regime.

The local training windows can be prescribed manually by providing the three local ROM identifiers, or generated automatically from the FOM angular-momentum history. In the automatic workflow, the first downward crossing of the pre-to-transition threshold and the first downward crossing of the transition-to-post threshold are detected in the FOM diagnostic data. A padding time is then added around the crossings so that the transition ROM is trained on snapshots slightly before and after the reversal event. This produces three overlapping windows: one for the initial circulation, one for the reversal, and one for the final circulation. The same FOM crossing times are then stored as the fixed switching schedule. The switched ROM therefore remains data-driven, but its online prediction uses a prescribed sequence of local reduced coordinates rather than a state-dependent ROM-triggered switch.

Chapter 5

Numerical Experiments

5.1 Reported error measures

The numerical results compare ROM trajectories with a reference FOM trajectory at common output times. The main state-space metric is the relative reconstruction error

$$e_x(t_k) = \frac{\|\boldsymbol{x}_{\mathrm{ROM}}(t_k) - \boldsymbol{x}_{\mathrm{FOM}}(t_k)\|_2}{\|\boldsymbol{x}_{\mathrm{FOM}}(t_k)\|_2}. \tag{5.1}$$

For Rayleigh–Bénard convection, $\boldsymbol{x}$ is the coupled velocity–temperature state. Scalar diagnostics, such as kinetic energy, normalized angular momentum and Nusselt number, are compared using a relative root-mean-square (RMS) error,

$$e_y^{\mathrm{RMS}} = \frac{\left(\frac{1}{K}\sum_{k=1}^{K}\left(y_{\mathrm{ROM}}(t_k) - y_{\mathrm{FOM}}(t_k)\right)^2\right)^{1/2}}{\left(\frac{1}{K}\sum_{k=1}^{K} y_{\mathrm{FOM}}(t_k)^2\right)^{1/2}}. \tag{5.2}$$

Here RMS denotes root-mean-square. This metric reports the typical diagnostic error over the comparison interval, normalized by the typical size of the corresponding FOM diagnostic.

Reduced operators are compared with absolute and relative Frobenius errors.[1] For a reduced operator O and reference operator O_{ref}, the difference is $\Delta O = O - O_{\mathrm{ref}}$, the absolute operator error is $\|\Delta O\|_F$, and the relative error reported in the tables is

$$e_O^F = \frac{\|\Delta O\|_F}{\|O_{\mathrm{ref}}\|_F}. \tag{5.3}$$

For vector-valued affine terms, the same notation is used and the Frobenius norm reduces to the Euclidean norm.

When runtime values are reported, they are the timings stored by the present implementation and should be interpreted as run timings for the current hardware and code path, not as architecture-independent complexity constants.

5.2 Two-dimensional dipole–wall interaction

The first test case considered in this thesis is the two-dimensional dipole–wall interaction. This flow has been used as a benchmark for unsteady incompressible flow solvers because,

[1] For a matrix B, the Frobenius norm is $\|B\|_F = \left(\sum_{i,j} B_{ij}^2\right)^{1/2}$.

despite its simple initial condition, it generates thin boundary layers, secondary vortices and small-scale vorticity filaments after impact with the wall [13, 22]. Therefore, it provides a suitable test case both for validating the full-order model (FOM) and for studying the performance of reduced-order models (ROMs) in a flow with strongly localized vortical structures.

Only the normal collision at Reynolds number

$$Re = 2500$$

is considered in this work. The computational domain is the square

$$\Omega = [-1, 1] \times [-1, 1],$$

with no-slip boundary conditions imposed on all walls,

$$\boldsymbol{u} = \boldsymbol{0} \qquad \text{on } \partial\Omega.$$

The initial condition consists of two counter-rotating vortices centred at

$$(x_1, y_1) = (0, 0.1), \qquad (x_2, y_2) = (0, -0.1),$$

so that the resulting dipole propagates towards the right wall, located at $x = 1$. Following [13, 22], the initial velocity field is prescribed as

$$u(x, y, 0) = \frac{|\omega_e|}{2}\left[-(y - y_1)\exp\left(-\left(\frac{r_1}{r_0}\right)^2\right) + (y - y_2)\exp\left(-\left(\frac{r_2}{r_0}\right)^2\right)\right], \tag{5.4}$$

$$v(x, y, 0) = \frac{|\omega_e|}{2}\left[(x - x_1)\exp\left(-\left(\frac{r_1}{r_0}\right)^2\right) - (x - x_2)\exp\left(-\left(\frac{r_2}{r_0}\right)^2\right)\right], \tag{5.5}$$

$$p(x, y, 0) = 0, \tag{5.6}$$

where

$$r_i^2 = (x - x_i)^2 + (y - y_i)^2, \qquad i = 1, 2, \qquad r_0 = 0.1.$$

The parameter $|\omega_e|$ is chosen such that the initial total kinetic energy is equal to

$$K(0) = \frac{1}{2}\int_\Omega |\boldsymbol{u}(x, y, 0)|^2 \, d\Omega = 2.$$

5.2.1 Validation of the FOM implementation

Before using the FOM data to construct and train ROMs, the numerical implementation is validated against the benchmark behaviour reported in [13, 22]. Since this problem is particularly sensitive to under-resolution near the no-slip wall, the validation is not based only on global quantities, but also on the local structure of the vorticity field after the primary wall collision.

The vorticity and enstrophy diagnostics are those defined in Equations (2.26) and (2.28). Figure 5.1 first shows the qualitative evolution of the validated FOM trajectory. The initially compact vortex dipole travels towards the right wall, generates a thin boundary layer during impact, and then produces secondary vortices and stretched vorticity filaments.

For the normal dipole–wall collision at $Re = 2500$, the reference simulations report two main enstrophy peaks, corresponding approximately to the first and second wall collisions of the primary vortices. These occur at $t \approx 0.33$ and $t \approx 0.61$, respectively [13].

A grid-convergence study has been performed using combinations of the relevant numerical parameters: the grid stretching factor in the x-direction[2], the Courant–Friedrichs–

[2] The grid stretching factor is the parameter used to generate a non-uniform spatial grid. It controls how strongly grid points are clustered near selected regions of the domain. In this test case, stretching in the x-direction is used to increase the resolution near the wall, where thin boundary layers and small-scale vorticity structures are formed.

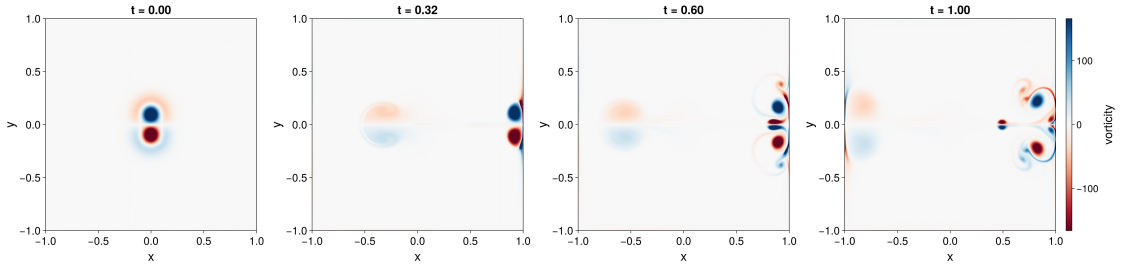


Figure 5.1: Vorticity snapshots from the 1536×1536 dipole–wall FOM at representative times. The panels show the initial dipole, the primary wall interaction, the second wall-interaction period, and the final state used for the grid-convergence comparison.

Lewy (CFL) number[3], and the number of cells in the grid. Figure 5.2 compares zoomed vorticity and pressure contours near the right wall at $t = 1$, in the region where the secondary vortex structures are most visible. The first column reproduces the benchmark contours reported by Knikker [22], while the remaining columns show the three FOM runs used in the grid-convergence study. The two coarser simulations do not fully reproduce the near-wall vortical structures: small vortices are either missing, displaced, or excessively diffused. In contrast, the simulation with 1536×1536 grid points captures the characteristic secondary vortex pair and the surrounding vorticity filaments observed in the benchmark result. The pressure contours show the same convergence trend in the near-wall pressure structure. Since pressure is defined only up to an additive constant, this comparison is based on the topology and location of the contours rather than on their absolute level. The 1536×1536 resolution is therefore selected as the converged FOM configuration used for the ROM experiments.

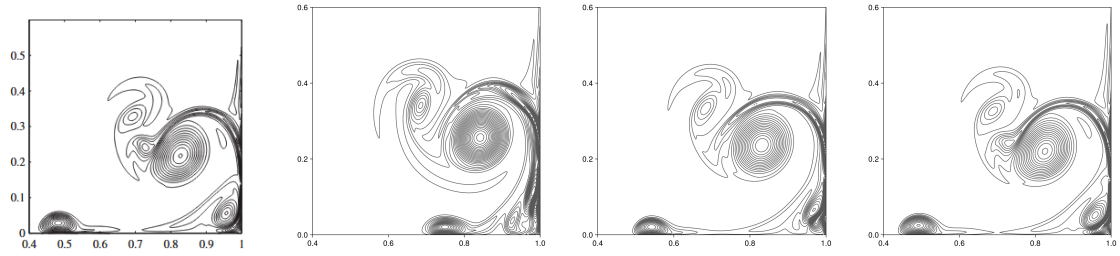
The enstrophy evolution in Figure 5.3 gives a complementary global comparison. Because enstrophy is integrated over the full domain, two simulations may have similar total enstrophy even if localized vortices are slightly displaced or diffused. For this reason, Figure 5.3b also shows the absolute enstrophy error with respect to the 1536×1536 solution. The absolute error is used here instead of a pointwise relative error because the purpose is to compare the magnitude and timing of the enstrophy deviations; a relative error would divide by the instantaneous reference enstrophy and can become visually dominated by times where this denominator is comparatively small rather than by the physically important peak errors. The comparison with the published benchmarks is made quantitatively in Table 5.1, which reports the two enstrophy peaks tabulated by Clercx and Bruneau [13].

For reference, Table 5.1 reports the benchmark enstrophy maxima from [13]. The values obtained with the finest FOM grid in this work are included for comparison. Since the 1536×1536 FOM stores snapshots every $\Delta t = 0.02$, the reported maxima are estimated by a local quadratic interpolation through the three stored enstrophy samples surrounding each peak.

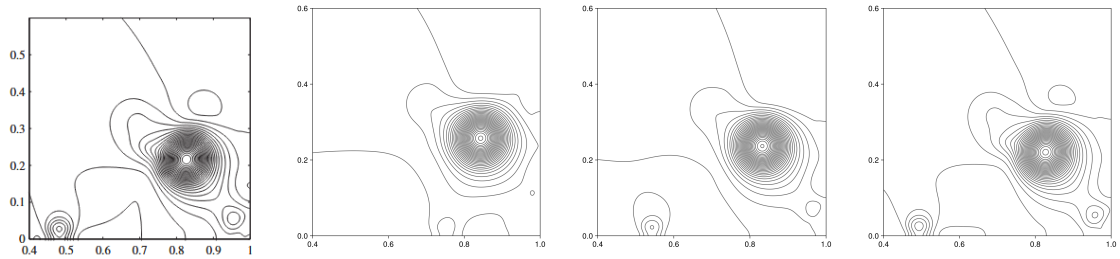
Table 5.1: Reference enstrophy maxima for the normal dipole–wall collision at $Re = 2500$, together with the corresponding values obtained in this work.

Source	t_1	$\mathscr{Z}(t_1)$	t_2	$\mathscr{Z}(t_2)$
Spectral benchmark [13]	0.3279	3313	0.6089	1418
Finite-difference benchmark [13]	0.3280	3270	0.6080	1408
This work, 1536×1536 FOM	0.3280	3277	0.6078	1407

[3]The CFL number is a dimensionless stability parameter that relates the time-step size to the spatial grid size and the local flow velocity. In general, smaller CFL numbers correspond to smaller time steps and therefore more stable, but more expensive, time integration [31].

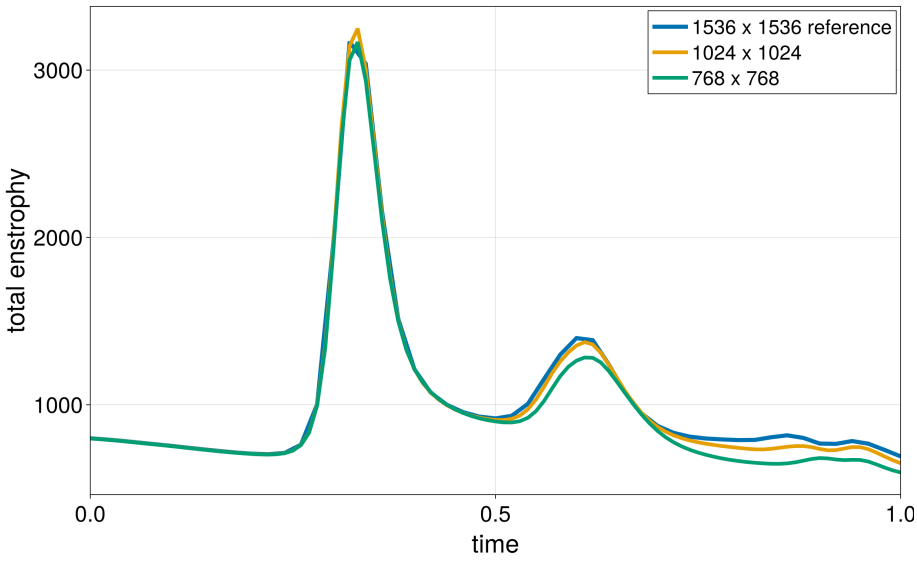


(a) Benchmark ω (b) 768^2 ω (c) 1024^2 ω (d) 1536^2 ω

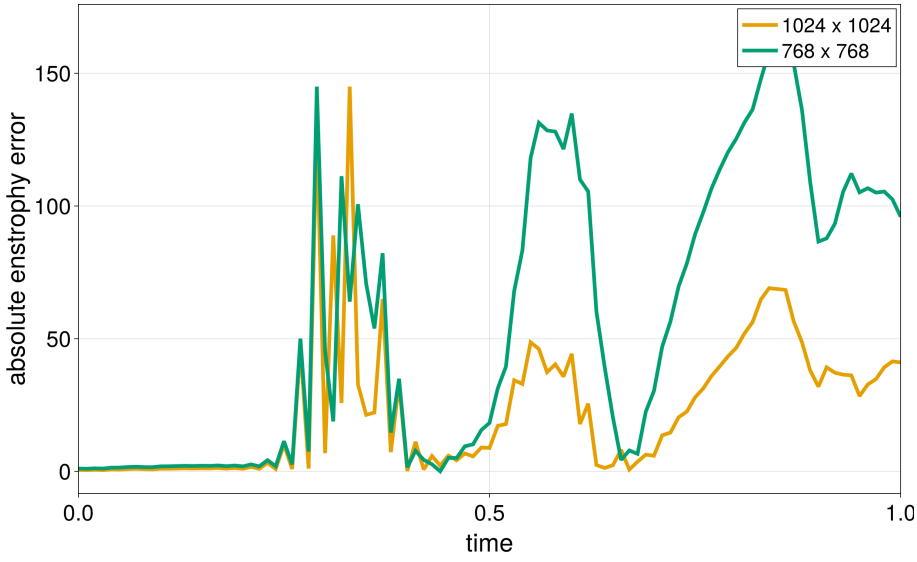


(e) Benchmark p (f) 768^2 p (g) 1024^2 p (h) 1536^2 p

Figure 5.2: Grid-convergence of the near-wall vorticity and pressure fields for the normal dipole–wall collision at $Re = 2500$ and $t = 1$. The first column shows the benchmark contours from [22]; the remaining columns show the FOM results with increasing grid resolution. The finest grid reproduces the benchmark secondary-vortex and pressure-contour structure and is used as the converged FOM reference.



(a) Total enstrophy.



(b) Error with respect to the reference solution.

Figure 5.3: Grid-convergence study for the normal dipole–wall collision at $Re = 2500$. Top: evolution of the total enstrophy $\mathscr{E}(t)$ for different grid resolutions. Bottom: absolute difference between the enstrophy obtained with non-converged grids and the reference solution.

Overall, the grid-refinement study confirms that the FOM reproduces the main benchmark features of the dipole–wall interaction at $Re = 2500$: the primary collision with the no-slip wall, the formation of secondary vortices, and the characteristic enstrophy peaks associated with the wall interactions. The 1536×1536 simulation is therefore taken as the reference full-order solution for the reduced-order modelling results presented below.

5.2.2 ROM reconstruction of the dipole–wall collision

The ROMs for the dipole–wall test case are trained on the validated 1536×1536 FOM trajectory described above. In contrast to the Rayleigh–Bénard case, no switched ROM is used here. The main purpose of this test is therefore to assess the exact OpInf construction in a single-basis setting: does it recover the intrusive Galerkin ROM to numerical precision, and how well does the resulting reduced trajectory reproduce the FOM wall-collision dynamics? State and diagnostic errors in the following comparisons use the measures defined in Section 5.1.

The primary comparison is between exact OpInf and the direct intrusive Galerkin ROM. Both use the same global POD basis, the same $r = 22$ reduced dimension selected by the $\eta = 0.99$ energy criterion, and the same quadratic reduced state space. The intrusive ROM is assembled by direct projection of the implemented diffusion and convection operators; the matrix-free details are given in Appendix A.1. Exact OpInf instead obtains the reduced operators from one-step FOM queries, as described in Chapter 4. A lower-rank exact OpInf ROM with $\eta = 0.95$ and a conventional trajectory OpInf ROM are then included as secondary comparisons. The trajectory OpInf run requests the same $\eta = 0.99$ POD basis, but the quadratic regression is limited by the 51 available trajectory snapshots. As discussed in Chapter 4, a quadratic trajectory-OpInf model of rank r requires

$$r + \frac{r(r+1)}{2}$$

independent feature samples before the reduced operators are identifiable from trajectory data. The $\eta = 0.99$ dipole basis has rank $r = 22$, which would require 275 feature samples. This is far more than the 51 available snapshots. Regularization could still force a rank-22 least-squares solve, but the missing 224 independent constraints would then be supplied by the regularization rather than by flow data. For this reason, the trajectory-OpInf model is fitted only at the largest identifiable quadratic rank, $r = 8$, for which $8 + 8 \cdot 9/2 = 44 \leq 51$. It therefore retains only 0.86663 of the POD energy. This distinction is essential when interpreting its accuracy and cost.

Table 5.2: Operator comparison between the exact OpInf and direct intrusive Galerkin ROMs for the dipole–wall case at $\eta = 0.99$.

Rank	$\|\Delta L\|_F$	Rel.	$\|\Delta Q\|_F$	Rel.
22	7.062×10^{-13}	2.295×10^{-13}	2.471×10^{-12}	1.769×10^{-14}

Table 5.2 is the main verification result for the dipole test case. The relative Frobenius error is 2.295×10^{-13} for the linear operator and 1.769×10^{-14} for the quadratic operator. Thus, for the same POD basis and rank, exact OpInf identifies the same reduced operators as the direct intrusive Galerkin projection up to floating-point round-off. This is stronger than obtaining a visually good trajectory fit: it checks that the exact OpInf snapshot-query construction reproduces the intrusive reduced model implemented in the code.

The accuracy of the same ROMs, measured against the 1536×1536 FOM, is summarized in Table 5.4.

Table 5.3 shows the online computational advantage of the reduced models, not a full workflow runtime comparison. The $\eta = 0.99$ exact OpInf and intrusive Galerkin ROMs ad-

Table 5.3: Online advance timing for the dipole–wall ROMs. The reference 1536×1536 FOM runtime stored in the metadata was 3285.58 s on the GPU; this is the timed full-order trajectory run after solver setup, including snapshot streaming. ROM timings report only the median reduced-coordinate advance time after a warm-up run, assuming that the reduced operators are already available. They are not end-to-end ROM runtimes and do not include offline basis construction, operator identification, full-state reconstruction or diagnostic postprocessing. Separate offline timing reruns for the exact OpInf cases are discussed below.

Method	Rank	Advance [s]	Speed-up
Exact OpInf, $\eta = 0.95$	13	0.06505	5.05×10^4
Exact OpInf, $\eta = 0.99$	22	0.21749	1.51×10^4
Intrusive, $\eta = 0.99$	22	0.12468	2.64×10^4
Trajectory OpInf	8	0.00488	6.74×10^5

Table 5.4: Dipole–wall ROM comparison against the 1536×1536 FOM. The retained energy is the actual POD energy retained after any rank truncation.

Method	Rank	Energy	Mean e_x	Max. e_x	Final e_x	e_K^{RMS}	$e_{\mathcal{Z}}^{\mathrm{RMS}}$	Final vort. err.
Exact OpInf, $\eta = 0.95$	13	0.95064	0.43372	0.75130	0.43960	0.04244	0.30343	0.64986
Exact OpInf, $\eta = 0.99$	22	0.99071	0.49750	1.06714	1.06640	0.01873	0.28216	1.70198
Intrusive, $\eta = 0.99$	22	0.99071	0.49750	1.06714	1.06640	0.01873	0.28216	1.70198
Trajectory OpInf	8	0.86663	0.50623	0.73208	0.47664	0.32749	0.59544	0.69950

vance in roughly 0.12–0.22 s once their operators are available, giving online cost ratios of order 10^4 relative to the stored FOM runtime. The corresponding one-time offline costs for the exact OpInf workflow were measured with additional no-plot and no-movie timing reruns: 230.52 s for $\eta = 0.95$ and 350.89 s for $\eta = 0.99$ on the CPU backend. These offline timings include snapshot loading, POD basis construction, rank-ensuring FOM queries, operator fitting, and reduced trajectory generation, but not figure or movie generation. They correspond to about 7% and 11% of the stored FOM runtime, respectively, and are much larger than the subsecond online advance times in Table 5.3. Thus, the reported speed-ups should be interpreted as online speed-ups; the model-building cost must be included when assessing the end-to-end cost for a single trajectory. The trajectory OpInf ROM is faster because it uses only rank 8, but this speed should be interpreted together with the accuracy results in Table 5.4: the lower cost is obtained by retaining substantially less POD energy and by damping the enstrophy dynamics. The timing table therefore supports the main trade-off in this test case. Exact OpInf is not the cheapest reduced model, but it gives the high-rank intrusive ROM at essentially the same online cost as the intrusive Galerkin ROM and with orders-of-magnitude less online work than advancing the FOM.

The same conclusion holds at trajectory level. Figure 5.4 shows that the exact OpInf and intrusive Galerkin $\eta = 0.99$ curves lie on top of each other, and Table 5.4 reports identical state-error and diagnostic statistics to the shown precision. The exact OpInf ROM therefore reconstructs the intrusive ROM trajectory as well as the operators. Relative to the FOM, the reduced trajectory is accurate during the early translation of the dipole, but the error grows after the primary wall collision. By $t = 1$, the state error is 1.06640. This should be interpreted as a limitation of the global projection-based ROM for the wall-collision dynamics, not as a failure of exact OpInf to recover the intrusive ROM.

The time-dependent diagnostics give the clearest comparison with trajectory OpInf. In Figure 5.5, the $\eta = 0.99$ exact OpInf ROM follows the FOM kinetic-energy decay closely over the full interval; its relative kinetic-energy RMS error is 0.01873, compared with 0.32749 for trajectory OpInf. The enstrophy history is harder because it is dominated by thin vorticity layers generated at the wall. Even here, exact OpInf is better: the relative enstrophy RMS error is 0.28216, compared with 0.59544 for trajectory OpInf. The peak statistics in Table 5.5

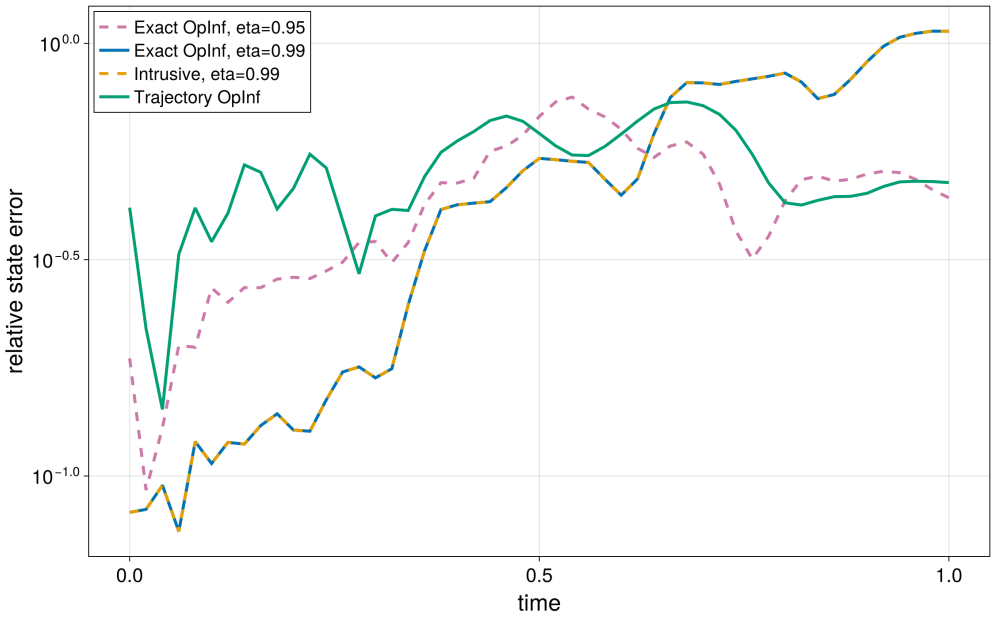


Figure 5.4: Relative state error for the dipole–wall ROMs. The exact OpInf and intrusive Galerkin $\eta = 0.99$ curves lie on top of each other, confirming that both runs advance the same reduced model.

Table 5.5: Additional dipole–wall diagnostic errors against the 1536×1536 FOM. The peak errors refer to the first and second enstrophy maxima, estimated by local quadratic interpolation around the stored enstrophy maximum.

Method	$\lvert\Delta t_1\rvert$	Rel. $\lvert\Delta \mathscr{E}_1\rvert$	$\lvert\Delta t_2\rvert$	Rel. $\lvert\Delta \mathscr{E}_2\rvert$	Near-wall corr.	Near-wall $\mathscr{E}$ ratio
Exact OpInf, $\eta = 0.95$	0.01164	0.41863	0.14250	0.29783	0.73388	0.90010
Exact OpInf, $\eta = 0.99$	0.00609	0.21614	0.19221	0.04585	-0.05169	2.25053
Intrusive, $\eta = 0.99$	0.00609	0.21614	0.19221	0.04585	-0.05169	2.25053
Trajectory OpInf	0.00823	0.61603	0.10604	0.68753	0.66366	0.42975

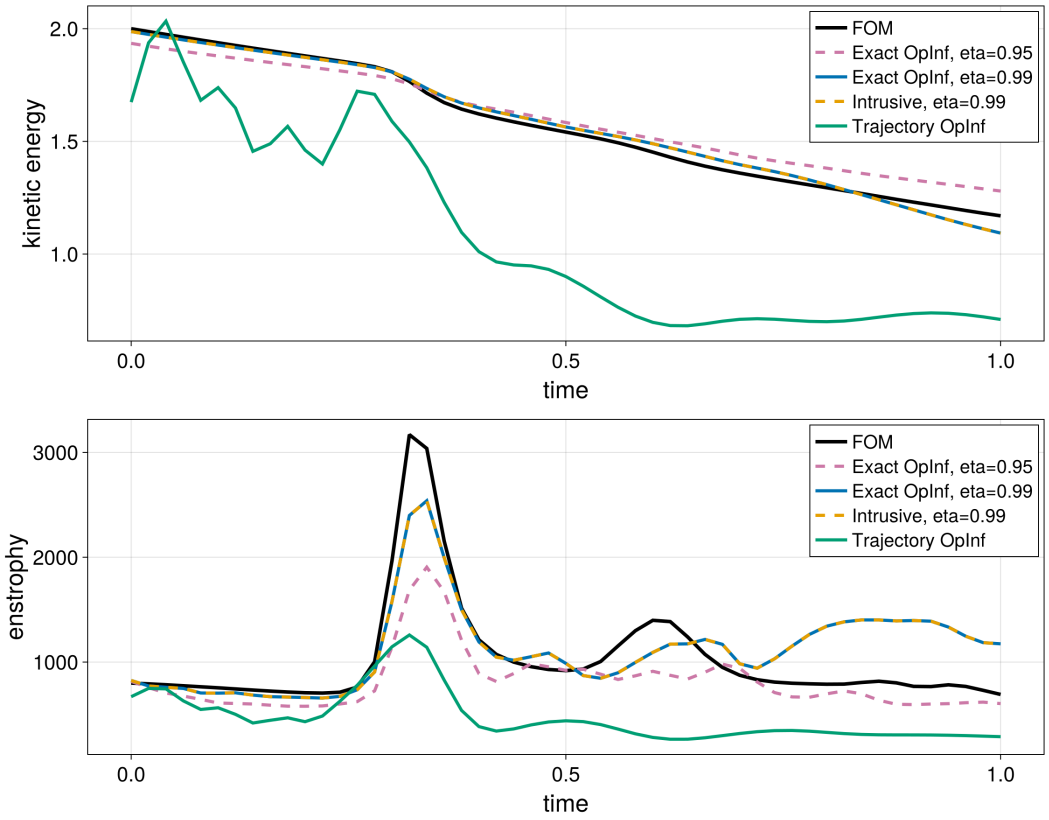


Figure 5.5: Kinetic-energy and enstrophy evolution for the dipole–wall FOM and ROMs. Exact OpInf and intrusive Galerkin at $\eta = 0.99$ overlap, while the trajectory OpInf ROM loses a large fraction of the energetic and vortical content after the first wall interaction.

show the same pattern. Exact OpInf underpredicts the first enstrophy peak by 21.614% and the second by only 4.585%, whereas trajectory OpInf underpredicts them by 61.603% and 68.753%, respectively.

This is the expected outcome for this comparison. Exact OpInf does not estimate the reduced operators from noisy finite-difference derivatives or solve an underdetermined trajectory regression. It queries the FOM one-step map in the same quadratic feature space as the intrusive Galerkin ROM, using deliberately chosen rank-ensuring reduced states rather than the limited set of states visited by one physical trajectory. Consequently, the $\eta = 0.99$, $r = 22$ exact OpInf operator fit is unregularized and identifiable. The conventional trajectory OpInf run cannot make the same 0.99-energy fit from the available data: at $r = 22$, the quadratic feature matrix would have 275 rows but only 51 columns. The reported trajectory-OpInf result is therefore necessarily a lower-rank comparison if the fitted operators are to be constrained by the data rather than by regularization. This lower-rank, regularized model gives a smoother trajectory and a smaller final state error than the $\eta = 0.99$ exact ROM, but the physical diagnostics show that this is not a better FOM surrogate: it loses kinetic energy too quickly, strongly damps the enstrophy peaks, and retains only 42.975% of the FOM near-wall enstrophy at the final time.

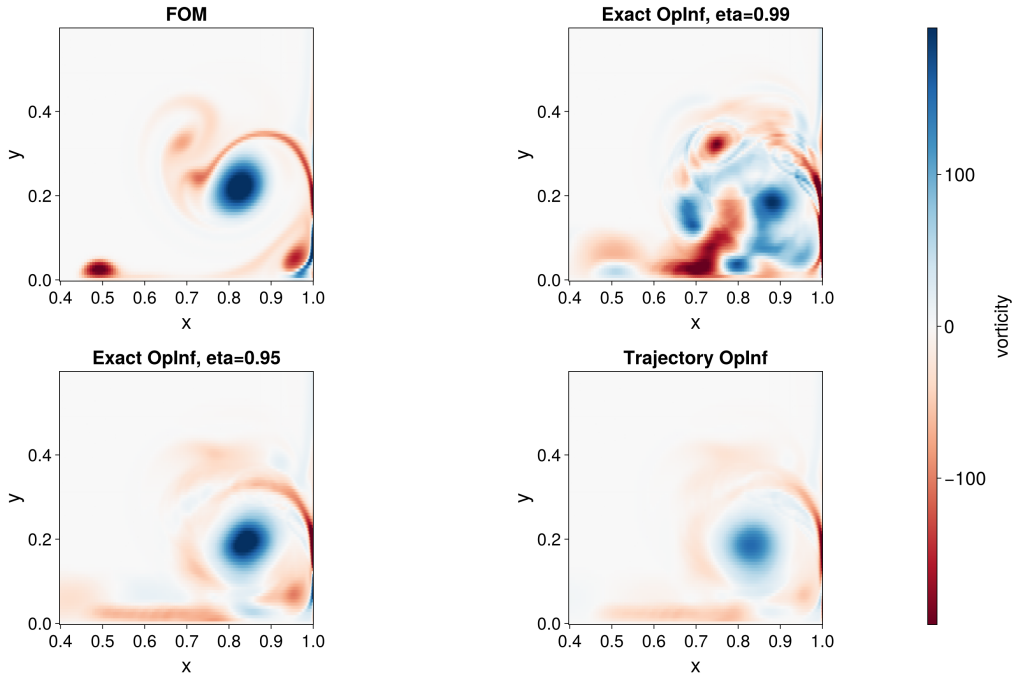


Figure 5.6: Final-time vorticity near the right wall for the FOM and selected ROMs. The $\eta = 0.99$ exact OpInf and intrusive Galerkin ROMs are identical at this scale, so only the exact OpInf field is shown. All panels use the shared robust colour range shown by the colour bar; values outside this range are clipped so that the near-wall vortical structures remain visible. Amplitude errors are quantified in Table 5.4.

Figure 5.6 explains where the FOM error comes from. The FOM contains a localized secondary vortex structure and a thin vorticity layer close to the wall. The $\eta = 0.99$ exact OpInf ROM follows the broad wall-layer location and gives the best global kinetic-energy RMS error (0.01873), but the final vorticity field contains oscillatory near-wall patches that are not present in the FOM. This gives a relative final-vorticity error of 1.70198. The lower-rank $\eta = 0.95$ exact OpInf ROM is smoother and has a lower final vorticity error, 0.64986, but it under-resolves the sharper vorticity content and gives larger enstrophy-peak errors. The

trajectory OpInf ROM is also smooth and has a moderate final state error, but because it is forced down to rank 8, it loses too much energetic content: its kinetic-energy and enstrophy RMS errors are 0.32749 and 0.59544, both substantially worse than the $\eta = 0.99$ exact OpInf ROM.

5.2.3 Discussion

The dipole–wall results support two main conclusions. First, exact OpInf succeeds at its central objective: it reproduces the intrusive Galerkin operators and the intrusive ROM trajectory to the reported precision in a two-dimensional incompressible flow problem. This verifies the operator-query construction independently of whether the resulting global ROM is an ideal surrogate for the FOM.

Second, the remaining discrepancy with the FOM is mainly a limitation of the global POD representation. The dipole–wall collision contains localized boundary-layer vorticity and secondary vortices, while a global POD basis orders modes by total kinetic-energy content rather than by their importance for near-wall vortex formation. Exact OpInf can recover the intrusive global ROM, but it cannot remove the truncation and closure limitations of that ROM. A time-local reduced basis, similar in spirit to the scheduled switched ROM used later for Rayleigh–Bénard convection, would therefore be a natural next step for this test case. A space-local basis could also be useful because the largest errors are concentrated near the wall; the space-local POD idea is discussed in Appendix C.1. Both extensions are outside the final comparison used in this thesis.

5.3 Rayleigh–Bénard Convection

The second test case considered in this thesis is two-dimensional Rayleigh–Bénard convection. This problem describes the motion of a fluid layer heated from below and cooled from above. The imposed temperature difference creates an unstable density stratification: warmer fluid near the lower wall becomes lighter and tends to rise, whereas colder fluid near the upper wall becomes heavier and tends to sink. The competition between buoyancy, viscous diffusion and thermal diffusion gives rise to convective rolls, thermal boundary layers and plume-like temperature structures [42, 20].

The computational domain is the square cavity

$$\Omega = [0,1] \times [0,1].$$

No-slip boundary conditions are imposed on all walls,

$$\boldsymbol{u} = \boldsymbol{0} \qquad \text{on } \partial\Omega.$$

The lower wall is kept at the hot temperature and the upper wall at the cold temperature,

$$T(x,0,t) = 1, \qquad T(x,1,t) = 0,$$

while the vertical side walls are thermally insulated,

$$\frac{\partial T}{\partial x}(0,y,t) = 0, \qquad \frac{\partial T}{\partial x}(1,y,t) = 0.$$

The continuous equations are the incompressible Boussinesq full-order model stated in Section 3.1. In this setup, the reference temperature and thermal expansion coefficient are absorbed into the imposed temperature scale, so the buoyancy contribution is represented by $gT\boldsymbol{e}_y$. In some formulations of the heat equation, an additional source term is included to account for viscous dissipation, which converts kinetic energy lost through viscosity into

internal energy and therefore leads to a fully energy-consistent formulation [16, 33]. In the present work, however, this term is omitted in order to reproduce the classical Rayleigh–Bénard convection setting without internal heat generation. A more detailed discussion of this modelling choice is provided in Appendix B.

In the present setup,

$$\nu = \kappa = 2.5 \times 10^{-4}, \qquad g = 1,$$

so that, with cavity height $H = 1$ and temperature difference $\Delta T = 1$, the Rayleigh number is

$$Ra = \frac{g \Delta T H^3}{\nu \kappa} = 1.6 \times 10^7,$$

and the Prandtl number is

$$Pr = \frac{\nu}{\kappa} = 1.$$

The initial velocity is zero,

$$\boldsymbol{u}(x, y, 0) = \mathbf{0},$$

and the temperature is initialized close to the mean temperature, with a small horizontal perturbation,

$$T(x, y, 0) = \frac{1}{2} + 0.01 \max(\sin(20\pi x), 0).$$

This perturbation breaks the symmetry of the initially motionless state and triggers the development of buoyancy-driven convection.

To make the subsequent grid and ROM discussion easier to interpret, Figure 5.7 shows four representative snapshots from the 256×256, $t = 150$ FOM trajectory used later for the switched-ROM experiments. The figure is included here as a qualitative visualization of the Rayleigh–Bénard dynamics; the spatial-resolution limitations of this trajectory are assessed in Section 5.3.1. The temperature field shows the evolution of thermal plumes and the large-scale roll, while the velocity arrows indicate the circulation direction.

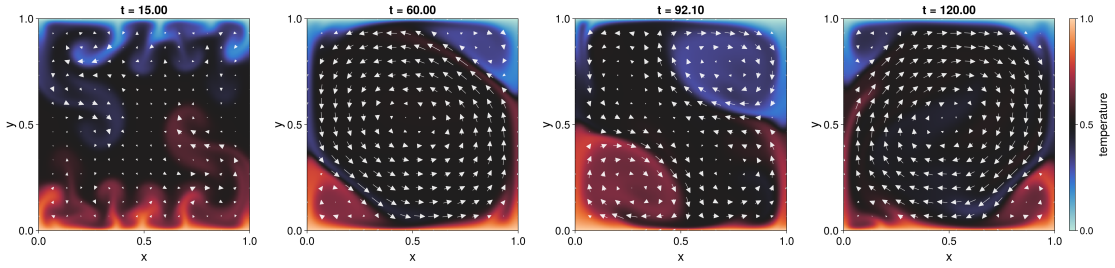


Figure 5.7: Temperature and velocity snapshots from the 256×256 Rayleigh–Bénard FOM used for the switched-ROM study. From left to right the panels show an early transient ($t = 15.00$), the pre-reversal rotating state ($t = 60.00$), the circulation reversal ($t = 92.10$), and the post-reversal rotating state ($t = 120.00$). The arrows show the cell-centred velocity field, downsampled for readability and scaled with one common factor across all panels.

5.3.1 FOM grid convergence

Before discussing the ROMs, it is important to clarify the resolution of the Rayleigh–Bénard reference data. Using a semi-automated experimental setup based on configuration files, several combinations of final time, grid size, snapshot density, dissipation-output settings and ROM parameters were tested. The most relevant FOM set for the present ROM study consists of four runs with the same physical parameters, $\nu = \kappa = 2.5 \times 10^{-4}$, $Pr = 1$, $Ra = 1.6 \times 10^7$, $g = 1$, CFL number 0.67, the same initial condition, and final time $t = 150$. The

only considered difference is the spatial grid in order to study grid convergence: 256×256, 384×384, 512×512, and 768×768.

Rayleigh–Bénard convection is particularly sensitive to grid resolution because heat transfer and momentum transport are controlled by thin thermal and kinetic boundary layers. These boundary layers affect the global quantities most often used to characterize the flow, including the Nusselt number and the strength of the large-scale circulation [2]. For this reason, Shishkina et al. formulate explicit boundary-layer resolution criteria for turbulent thermal convection [35]. The same type of diagnostic comparison is also useful in ROM studies: Nusselt number, kinetic energy, and circulation measures indicate whether the reduced dynamics remain close to the FOM; see for example the Rayleigh–Bénard ROM study of Chand, Rosenberger and Sanderse [11].

Figure 5.8 compares the four FOMs through kinetic energy, normalized angular momentum and volume-averaged Nusselt number. Table 5.6 reports scalar convergence statistics. The Nusselt means shown on the table are computed over $t \in [100, 150]$, after the initial transient.

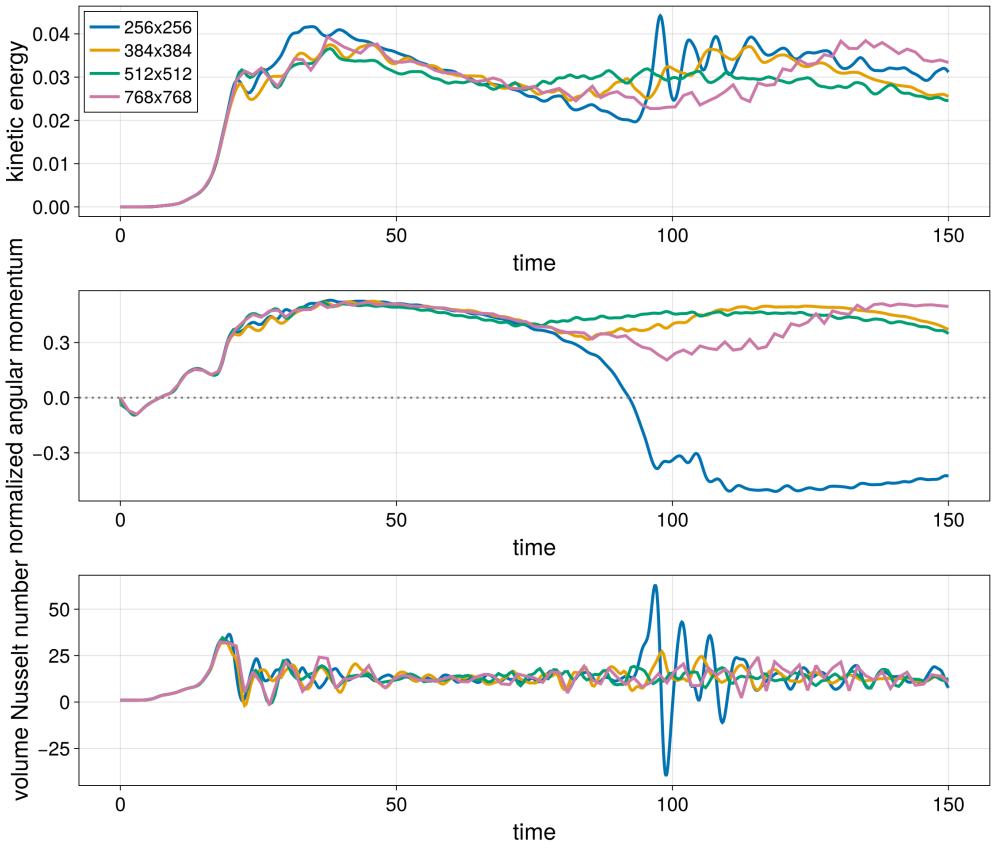


Figure 5.8: Grid comparison for the Rayleigh–Bénard FOMs with $Ra = 1.6 \times 10^7$ and $Pr = 1$. The 384×384, 512×512, and 768×768 trajectories all avoid the late circulation reversal seen on the 256×256 grid, although their pointwise diagnostics are not monotone with grid refinement.

The 384×384, 512×512, and 768×768 FOMs show the same qualitative large-scale behaviour. In all three runs, the normalized angular momentum remains positive at $t = 150$, whereas the 256×256 run undergoes a late circulation reversal and ends at $\ell_z = -0.426421$. This is the strongest indication that the 256×256 grid is not in the same grid-consistent regime as the finer simulations. Nevertheless, this reversal could be expected in a long

Table 5.6: FOM grid-convergence statistics for the Rayleigh–Bénard setup. Relative RMS errors are measured against the 768×768 run on the 101 common saved times. The Nusselt values are time averages over $t \in [100, 150]$.

Grid	Snapshots	Final K	Final ℓ_z	$\overline{Nu_b}$	$\overline{Nu_t}$	$\overline{Nu_V}$	e_K vs. 768	e_{ℓ_z} vs. 768	e_{Nu_V} vs. 768
256^2	1000	0.031151	-0.426421	13.855901	13.938794	14.089701	0.17592	1.30221	0.66096
384^2	500	0.025519	0.370162	13.911821	14.059059	13.483362	0.15791	0.21276	0.36331
512^2	300	0.024572	0.348220	13.504482	13.469815	13.127894	0.16223	0.23857	0.30927
768^2	100	0.033397	0.496719	13.815987	13.763690	14.309907	0.00000	0.00000	0.00000

enough trajectory, and the 256×256 run is therefore still useful for testing whether a reduced model can reproduce a supplied trajectory with a regime change.

The finer-grid comparison is not perfectly monotone. For example, the final kinetic energy of the 768×768 run is 0.033397, larger than the 384×384 and 512×512 values, 0.025519 and 0.024572. The volume-Nusselt root-mean-square (RMS) error against the 768×768 run is 0.36331 for 384×384 and 0.30927 for 512×512, so this diagnostic does improve from 384^2 to 512^2, but the pointwise kinetic-energy and angular-momentum differences remain visible. For this high-Rayleigh-number case, this means that grid refinement does not produce a simple monotone convergence of every time trace, even though the finer grids give a more consistent picture of the large-scale regime and heat-transfer statistics.

The grid study therefore separates two issues. The finer simulations are the appropriate evidence for spatial-resolution sensitivity, while the ROM study below is a reduced-modelling experiment performed on one selected FOM trajectory. That trajectory is introduced next.

5.3.2 Reference FOM and local training windows

The 256×256 FOM is used to test the ROMs in a deliberately challenging setting: the trajectory contains a large-scale circulation reversal, so the reduced model must reproduce not only smooth diagnostic evolution but also a clear regime change. This makes the run especially relevant for assessing the scheduled switched ROM strategy introduced in Section 4.4. The selected trajectory uses $\nu = \kappa = 2.5 \times 10^{-4}$, a 256×256 uniform grid, CFL number 0.67, final time $t = 150$, and stores 1001 snapshots including the initial condition. This is the trajectory visualized in Figure 5.7.

As shown in the grid study above, this 256×256 run is not used as a grid-converged reference for the physical Rayleigh–Bénard problem at $Ra = 1.6 \times 10^7$. The ROM results should therefore be read as reduced-modelling results relative to this chosen FOM trajectory: can a non-intrusive ROM reproduce its dynamics, diagnostics and reversal? They should not be read as a final grid-converged prediction of the physical flow.

Since both the velocity and the temperature are included in the state vector, the POD basis is built for the coupled velocity–temperature state. In all ROM comparisons below the reference FOM is kept fixed; only the reduced modelling strategy or the retained POD fluctuation energy is changed.

For this FOM, the scheduled switched ROMs use the angular-momentum-based procedure described in Section 4.4. The normalized angular-momentum history identifies the reversal interval, and the local windows are padded around the two detected transition times so that neighbouring models share snapshots near the change of circulation. On the stored snapshot grid this gives

$$\begin{aligned} \text{pre:} \quad & t \in [10.05, 85.65], \\ \text{transition:} \quad & t \in [77.70, 100.35], \\ \text{post:} \quad & t \in [92.40, 150.00]. \end{aligned}$$

The scheduled switching times obtained from the same FOM angular-momentum signal are

$$t_{\text{pre}\to\text{transition}} = 81.6636, \qquad t_{\text{transition}\to\text{post}} = 96.3869.$$

Because the stored trajectory uses the FOM snapshot grid, the switches are realized at

$$t = 81.75 \qquad \text{and} \qquad t = 96.45.$$

The active-model counts over the 934 predicted snapshots are therefore 478, 98, and 358 for the pre, transition and post models, respectively.

All state and diagnostic errors reported below are computed using the measures defined in Section 5.1.

5.3.3 Exact operator inference and the intrusive Galerkin reference

The first question is whether the exact operator-inference construction reproduces the intrusive Galerkin ROM when both methods use the same local bases and training windows. This is the central verification step for the exact OpInf implementation: if the snapshot-generation procedure is consistent, the inferred reduced operators should agree with the intrusive projected operators up to floating-point round-off.

Table 5.7 compares the local exact OpInf operators with the corresponding intrusive operators for the $\eta = 0.99$ scheduled setup. The constant term $\widetilde{c}$, linear operator $\widetilde{L}$, and quadratic operator $\widetilde{Q}$ are compared using Frobenius norms. All relative errors are between 10^{-15} and 10^{-13}. This confirms that, for the Rayleigh–Bénard test case, the exact OpInf procedure recovers the intrusive reduced operators to machine precision, even though the FOM is a coupled velocity–temperature system and the basis is centered around a nonzero mean state.

This agreement is an implementation verification rather than an independent physical conclusion. The intrusive reference used here is the sampled Galerkin projection of the implemented semi-discrete Boussinesq right-hand side, and exact OpInf queries the same right-hand side through Euler-consistent one-step samples. Under the assumptions stated in Chapter 4, this setup should recover the same affine-quadratic reduced map. The Rayleigh–Bénard implementation details and the limits of this interpretation are given in Appendix A.2.

Table 5.7: Operator comparison between local exact OpInf ROMs and local intrusive Galerkin ROMs for the Rayleigh–Bénard scheduled switched setup with retained fluctuation energy $\eta = 0.99$.

Regime	Rank	$\|\Delta c\|_F$	Rel.	$\|\Delta L\|_F$	Rel.	$\|\Delta Q\|_F$	Rel.
Pre	72	2.044×10^{-16}	8.050×10^{-15}	5.440×10^{-13}	2.262×10^{-14}	2.137×10^{-12}	7.266×10^{-15}
Transition	22	5.331×10^{-17}	1.539×10^{-15}	4.062×10^{-14}	8.148×10^{-15}	9.838×10^{-14}	1.612×10^{-15}
Post	73	1.231×10^{-15}	6.095×10^{-14}	5.753×10^{-13}	1.900×10^{-14}	2.294×10^{-12}	7.893×10^{-15}

The trajectory-level comparison in Table 5.8 is consistent with this operator result. The exact OpInf scheduled ROM and the intrusive scheduled ROM have the same local ranks, the same switching times, and identical reported error statistics to the precision shown. The mean relative state error is 0.16034, the median is 0.15546, and the final-time error is 0.19329. The relative RMS errors in kinetic energy and normalized angular momentum are 0.11116 and 0.07033, respectively. The intrusive run is slightly slower in this particular execution, but the difference is small compared with the advance cost of the $r_{\max} = 73$ switched system.

The speed-ups in Table 5.8 are more modest than in the dipole case because the $\eta = 0.99$ scheduled exact and intrusive ROMs have large local ranks $72/22/73$. Here $\eta = 0.99$ refers to retained centered POD fluctuation energy, not total physical energy. After subtracting the local mean state, the remaining reversal fluctuations are spread over many modes, which contributes to the higher ranks. Even so, their reduced-coordinate advance is about 3.5 times faster than the corresponding CPU FOM run. The trajectory OpInf run advances roughly 190 times faster than the FOM, but this is mainly a rank effect: it uses local ranks

Table 5.8: Rayleigh–Bénard method comparison for switched ROMs. All scheduled runs use the same switching times, $t = 81.75$ and $t = 96.45$. The trajectory OpInf run has lower local ranks because its affine-quadratic fit is limited by the number of available trajectory samples. The matching 256×256 FOM CPU rerun for $t \in [0, 150]$ took 578.113 s; this CPU FOM time is used for the FOM/advance ratios. The advance column reports the reduced-coordinate integration time used for the speed-up estimate; offline FOM generation, POD construction and operator identification are excluded. The offline cost of the scheduled exact OpInf case is discussed below.

Method	Ranks	Advance [s]	FOM/advance	Mean e_x	Max. e_x	Final e_x	e_K^{RMS}	$e_{\ell_z}^{\mathrm{RMS}}$
Exact OpInf, scheduled	72/22/73	162.200	3.56	0.16034	0.39050	0.19329	0.11116	0.07033
Intrusive, scheduled	72/22/73	164.354	3.52	0.16034	0.39050	0.19329	0.11116	0.07033
Trajectory OpInf, scheduled	30/15/26	3.039	190.23	0.19545	0.40303	0.31446	0.31024	0.14008

$30/15/26$, not the 0.99-energy ranks used by exact OpInf and intrusive Galerkin. The offline construction effort follows the opposite pattern. For the exact-OpInf scheduled setup, the three affine-quadratic local models require 2701, 276, and 2775 one-step feature queries before online prediction, respectively. These 5752 FOM-query samples, together with the local POD bases and dense operator solves, are part of the one-time model-building cost that is not included in the online timings in the table. An additional no-plot and no-movie timing rerun measured this scheduled exact OpInf construction workflow at 660.80 s on the CPU backend. This is about 1.14 times the matching CPU FOM runtime, so for a single trajectory the construction cost is comparable to, and slightly larger than, simply running the FOM once. It is also much larger than the 162.200 s online advance of the exact OpInf ROM. The reported speed-ups are therefore online speed-ups; the offline model-building cost must be amortized over reuse of the ROM, or included explicitly for a single end-to-end trajectory.

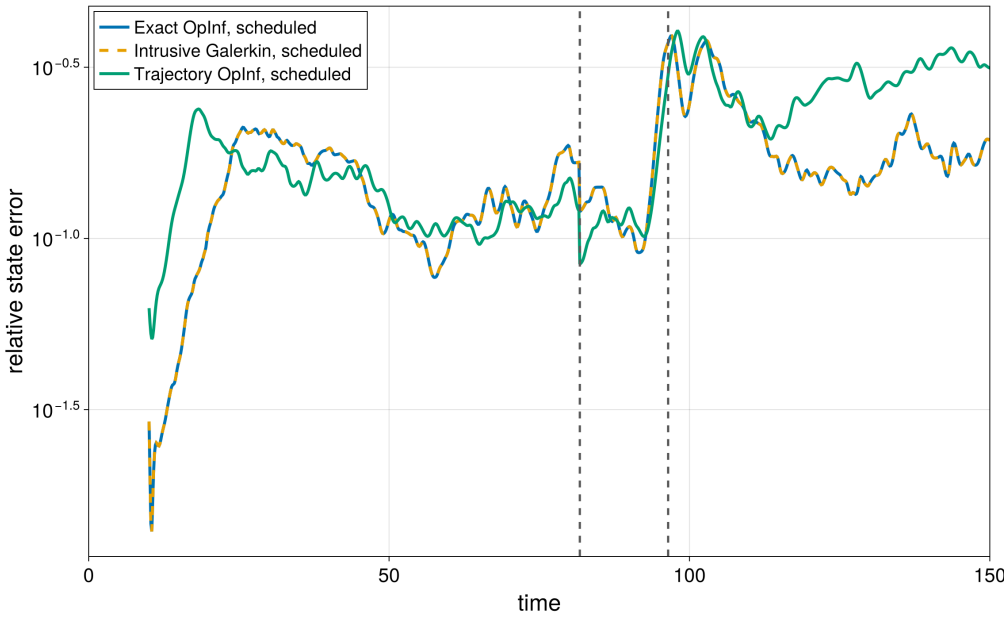
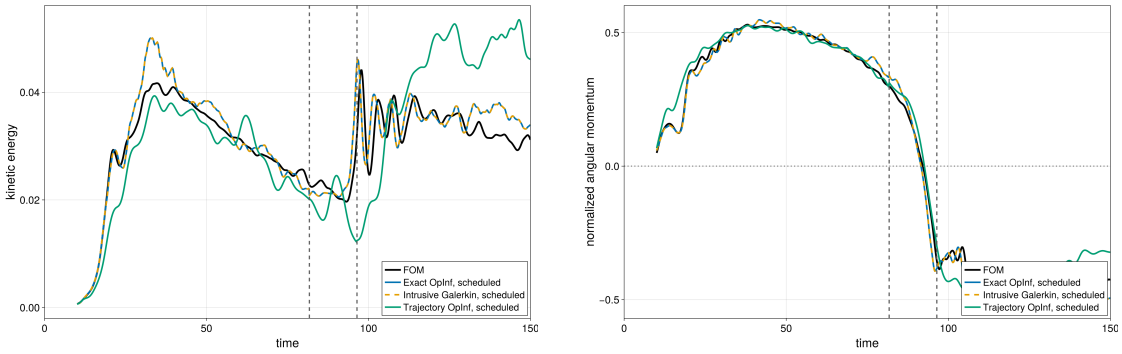


Figure 5.9: Relative state error for the scheduled exact OpInf, intrusive and trajectory OpInf switched ROMs at $\eta = 0.99$. The vertical dashed lines mark the scheduled switching times.

Figures 5.9 and 5.10 show the same comparison in time. The exact OpInf and intrusive Galerkin curves are visually indistinguishable, which is consistent with the operator agreement in Table 5.7. The trajectory OpInf ROM captures the reversal timing, but its lower local



(a) Kinetic energy.

(b) Normalized angular momentum.

Figure 5.10: Physical diagnostics for the scheduled $\eta = 0.99$ switched ROMs. Exact OpInf and intrusive Galerkin lie on top of each other, while trajectory OpInf captures the reversal timing but gives a visibly less accurate kinetic-energy evolution.

ranks lead to a different post-reversal error evolution and visibly less accurate kinetic-energy and angular-momentum histories. The source of this difference, and of the much larger online speed-up in Table 5.8, is the rank limitation described next.

The comparison with trajectory OpInf is more nuanced. The same 0.99 fluctuation-energy target is requested, but the trajectory-based affine-quadratic fit can only identify as many coefficients as are supported by the available trajectory samples. The implementation therefore caps the local ranks at $30/15/26$, instead of the $72/22/73$ ranks used by exact OpInf and the intrusive ROM. After this truncation, the retained fluctuation energy is 0.95382, 0.97861, and 0.93934 in the pre, transition, and post windows, respectively. Thus the trajectory OpInf run is not a true 0.99-energy comparison in all three windows. This rank reduction is the main reason for the much smaller online advance time, 3.039 seconds compared with 162.200 seconds for exact OpInf in Table 5.8: the trajectory OpInf run is advancing a much smaller quadratic ROM, not merely using a cheaper fitting procedure.

To separate the effect of the numerical method from the effect of rank, exact OpInf and intrusive Galerkin ROMs were also rerun with the same imposed local ranks $30/15/26$. These runs use the same local windows, the same retained fluctuation energies, and the same scheduled switching times as the trajectory OpInf run. The results are shown in Table 5.9 and Figure 5.11.

Table 5.9: Matched-rank Rayleigh–Bénard method comparison. All three switched ROMs use local ranks $30/15/26$, retained fluctuation energies $0.95382/0.97861/0.93934$, and scheduled switch times $t = 81.75$ and $t = 96.45$. The same CPU FOM baseline as in Table 5.8 is used for the FOM/advance ratios.

Method	Ranks	Advance [s]	FOM/advance	Mean e_x	Max. e_x	Final e_x	e_K^{RMS}	$e_{\ell_z}^{\mathrm{RMS}}$
Exact OpInf, matched ranks	30/15/26	5.820	99.33	0.66331	1.48308	0.54070	1.59386	0.54601
Intrusive, matched ranks	30/15/26	5.731	100.87	0.68522	1.48091	0.69427	1.62135	0.54377
Trajectory OpInf, matched ranks	30/15/26	3.039	190.23	0.19545	0.40303	0.31446	0.31024	0.14008

The matched-rank comparison confirms that most of the cost difference in Table 5.8 is caused by the reduced dimension. When exact OpInf and intrusive Galerkin are forced to use $r_{\max} = 30$, their online advance times drop to 5.820 and 5.731 seconds, respectively, close to the 3.039 seconds of trajectory OpInf. However, this cheaper projected model is no longer accurate for the reversal trajectory. The mean state error rises above 0.66, the maximum state error reaches about 1.48, and the kinetic energy RMS error is larger than 1.5. By contrast, trajectory OpInf gives a mean state error of 0.19545 at the same ranks. This lower

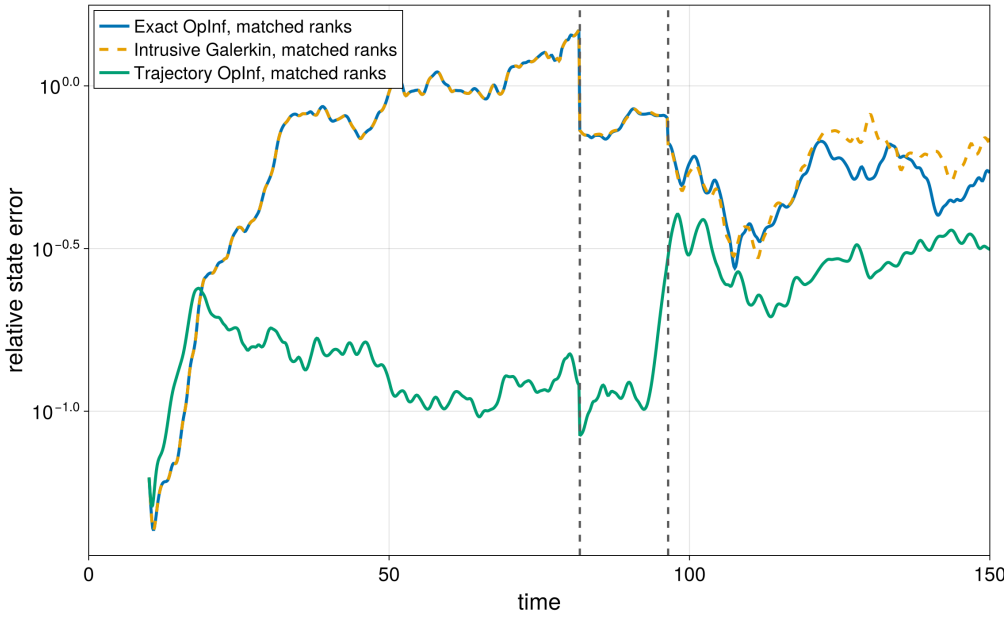


Figure 5.11: Relative state error for the matched-rank scheduled switched ROMs. The exact OpInf and intrusive Galerkin ROMs use the same reduced dimensions as trajectory OpInf, and the vertical dashed lines mark the scheduled switching times.

error is not surprising: trajectory OpInf fits the reduced operators directly to the projected FOM trajectory, so the regression can absorb part of the influence of discarded POD modes into the fitted coefficients. In contrast, the matched-rank exact OpInf and intrusive Galerkin ROMs are projected models of the resolved coordinates only; they do not include an additional closure term for the unresolved modes. At these low ranks, the trajectory-based fit therefore behaves like a closure-corrected reduced dynamics for the retained coordinates, whereas the projected exact OpInf and intrusive Galerkin systems suffer from unresolved truncation effects.

Relative to the full-rank exact OpInf run in Table 5.8, trajectory OpInf follows the same scheduled regime sequence and produces a mean relative state error of 0.19545, only moderately larger than 0.16034. However, the final-time error increases from 0.19329 to 0.31446, and the kinetic-energy RMS error increases from 0.11116 to 0.31024. Thus the conventional non-intrusive trajectory fit is a useful cheap surrogate for the large-scale transition, but it is less reliable than the full-rank exact OpInf or intrusive Galerkin ROM for reproducing the energetic dynamics of this buoyancy-driven flow.

As an additional sensitivity check, a threshold-switched exact OpInf run was also tested. Unlike the scheduled runs in Table 5.8, this variant switches using the ROM's own online angular-momentum threshold. Its first and second switches occur at $t = 85.20$ and $t = 99.15$, later than the scheduled switches. This increases the maximum relative state error from 0.39050 to 0.69010 and the angular-momentum RMS error from 0.07033 to 0.11400. This run is therefore relevant as a justification for using scheduled FOM-derived switching times, but it is not part of the main method comparison.

5.3.4 Influence of retained fluctuation energy

The next question is how much POD fluctuation energy must be retained for the switched ROM to reconstruct the reversal dynamics. Four scheduled exact OpInf switched ROMs are compared: $\eta = 0.90$, 0.95, 0.97, and 0.99. The same FOM trajectory, local time windows

and scheduled switch times are used in all cases. Therefore, the differences reported in Table 5.10 are caused by the selected local POD dimensions and by the resulting reduced operators.

Table 5.10: Effect of retained POD fluctuation energy on the scheduled exact OpInf switched ROM. The local ranks correspond to the pre, transition and post local bases. The same CPU FOM baseline as in Table 5.8 is used for the FOM/advance ratios.

η	Ranks	$r_{\max}$	Advance [s]	FOM/advance	Mean e_x	Max. e_x	e_K^{RMS}	$e_{\ell_z}^{\mathrm{RMS}}$
0.90	15/4/18	18	1.240	466.22	0.71502	2.19661	3.40696	0.40871
0.95	29/8/30	30	6.426	89.96	0.56716	1.55274	2.61146	0.37280
0.97	41/12/42	42	22.791	25.37	0.64769	1.43229	1.80481	0.57486
0.99	72/22/73	73	162.200	3.56	0.16034	0.39050	0.11116	0.07033

The same no-plot and no-movie timing procedure also shows how the one-time exact OpInf construction workflow grows with the retained fluctuation-energy threshold. These timings include local POD construction, exact one-step FOM queries, dense operator identification and switched-ROM trajectory generation, but exclude the original FOM simulation and diagnostic post-processing. Since the switched trajectory is also generated, they are conservative workflow timings rather than pure operator-fit times. The measured CPU wall-clock times were 109.14 s for $\eta = 0.90$, 121.09 s for $\eta = 0.95$, 187.74 s for $\eta = 0.97$, and 660.80 s for $\eta = 0.99$, the scheduled exact OpInf case discussed above.

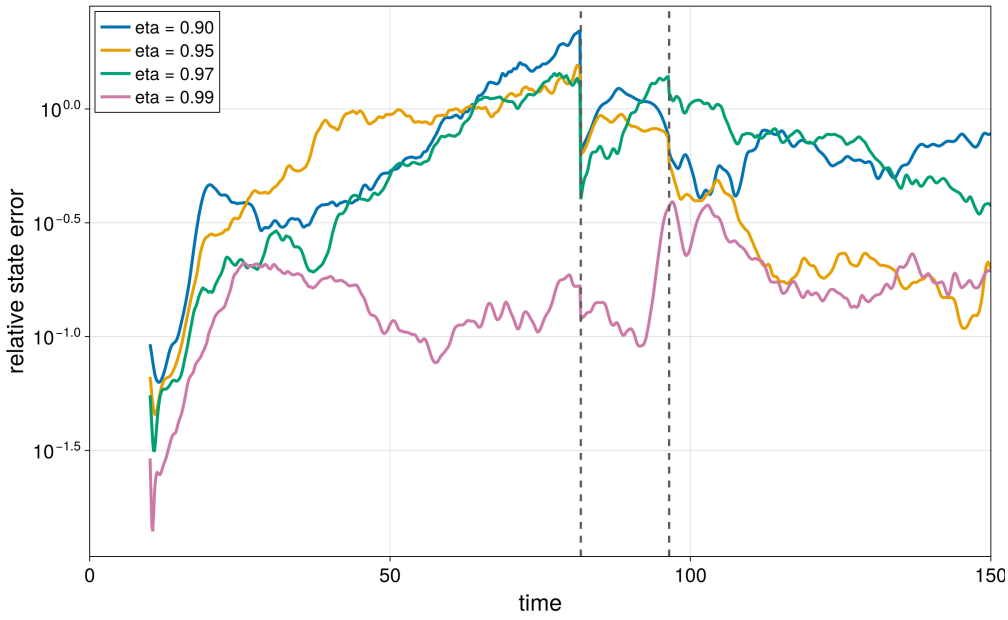
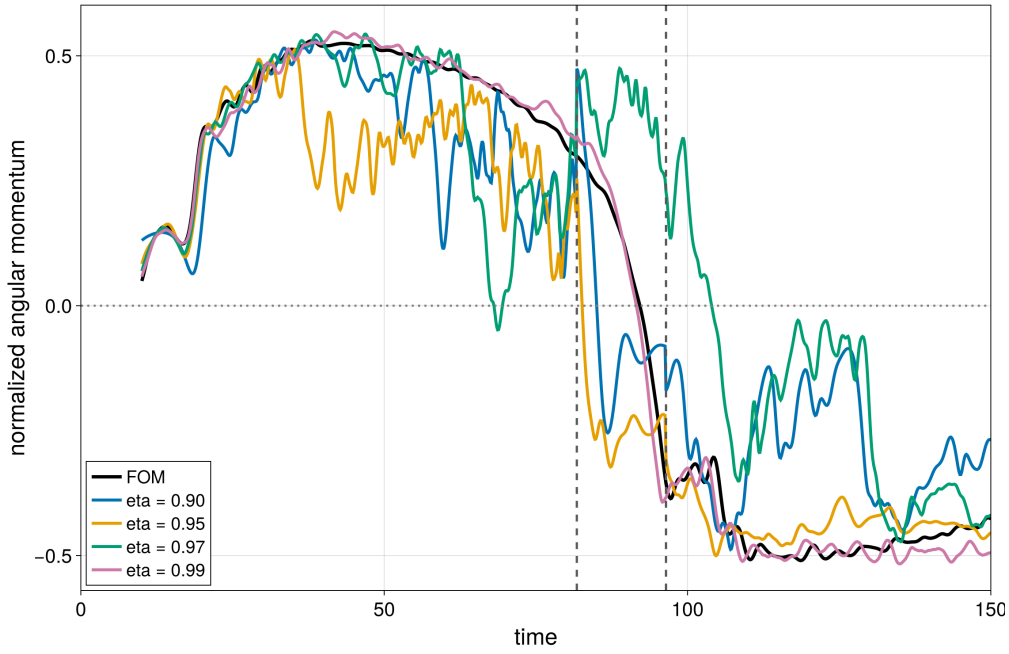
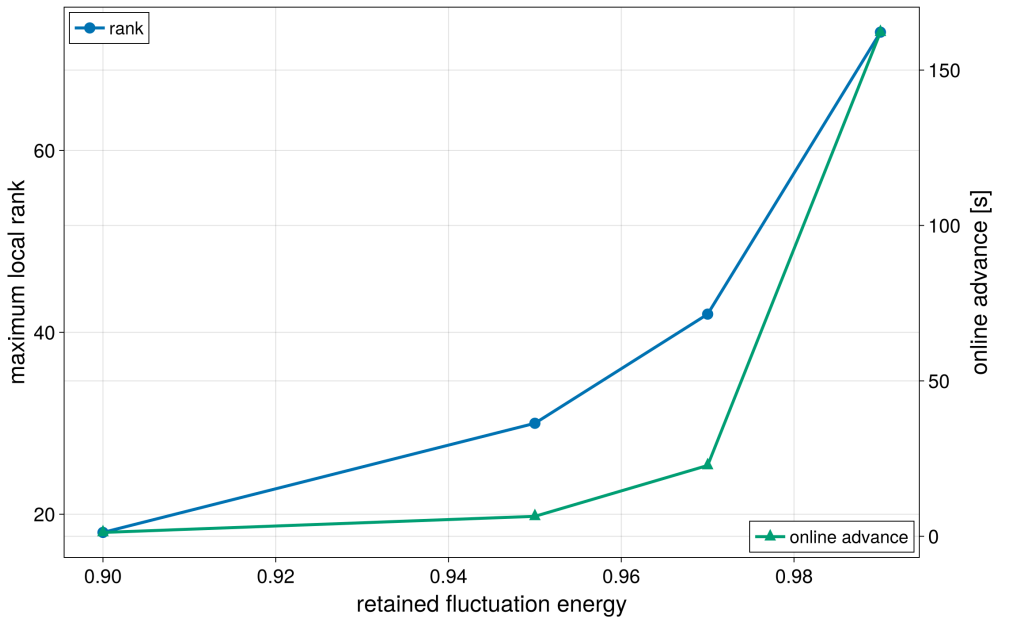


Figure 5.12: Relative state error for scheduled exact OpInf switched ROMs with different retained fluctuation-energy thresholds. The vertical dashed lines indicate the scheduled switch times.

Figure 5.13 complements the aggregate errors in Table 5.10. Panel (a) shows that the lower-energy ROMs do not track the angular-momentum reversal consistently, even when some scalar error measures improve. Panel (b) should be read as a trade-off plot: the blue curve uses the left axis for the maximum local rank, while the green curve uses the right axis for the online advance time. The offline construction timings are reported separately above. The sharp increase from $\eta = 0.97$ to $\eta = 0.99$ occurs because retaining the last few percent



(a) Normalized angular momentum.



(b) Rank and online advance scaling.

Figure 5.13: Effect of retained fluctuation energy on circulation reconstruction and online switched-ROM cost. The $\eta = 0.99$ ROM is much more expensive to advance, but it is the only sweep member that accurately follows the normalized angular momentum throughout the reversal and post-reversal regime.

of fluctuation energy increases the maximum local rank from 42 to 73; for a quadratic ROM, this rank increase strongly affects the online cost.

The results show that low-dimensional bases can be misleading for this problem. At $\eta = 0.90$, the maximum local rank is only 18, and the online advance takes 1.240 seconds. This is computationally attractive, but the mean relative state error is 0.71502, the maximum error exceeds 2, and the kinetic-energy RMS error is 3.40696. The ROM has too few modes to represent the coupled velocity–temperature structures involved in the reversal.

Increasing the threshold to $\eta = 0.95$ improves the mean state error to 0.56716 and lowers the angular-momentum RMS error to 0.37280, but the energetic dynamics are still poor: the kinetic-energy RMS error remains 2.61146. The $\eta = 0.97$ run is not uniformly better than $\eta = 0.95$. Although the maximum state error decreases from 1.55274 to 1.43229, the mean state error increases to 0.64769, and the angular-momentum error worsens to 0.57486. This non-monotonicity is important. Retaining more modes improves the projection space, but it also changes the inferred nonlinear dynamics and the projection jumps at the switch times. For a strongly transient reversal, intermediate ranks may contain enough extra degrees of freedom to change the dynamics without yet containing enough information to stabilize them.

The $\eta = 0.99$ model is qualitatively different. Its local ranks are 72/22/73, and the online advance time rises sharply to 162.200 seconds. This is about 131 times larger than the $\eta = 0.90$ advance time and about 25 times larger than the $\eta = 0.95$ advance time. However, it also reduces the mean relative state error to 0.16034, the maximum state error to 0.39050, the kinetic-energy RMS error to 0.11116, and the normalized-angular-momentum RMS error to 0.07033. The conclusion is therefore not simply that more modes are always better; rather, this Rayleigh–Bénard reversal requires a high fluctuation-energy threshold before the local ROMs contain enough information to reconstruct the coupled thermal and velocity dynamics.

5.3.5 Prediction beyond the training interval

The previous comparisons are primarily reconstruction tests: the ROMs are evaluated on the same time interval used to construct the local reduced bases and operators. To assess predictive capability more directly, an additional switched exact OpInf ROM is compared with a full-time FOM run on the same 256^2 grid and with the same physical parameters, but extended to $t = 300$. The ROM uses the same three local exact OpInf models trained from the $t = 150$ reference FOM: the pre-reversal model is trained on $t = 10.05, \ldots, 85.65$, the transition model on $t = 77.70, \ldots, 100.35$, and the post-reversal model on $t = 92.40, \ldots, 150.00$. Therefore, the interval $t > 150$ is a genuine extrapolation of the learned post-reversal dynamics. Consistently with the scheduled switched ROMs above, the FOM-derived switch times from the known training interval are kept fixed. The continuous crossing times $t_{\text{pre}\to\text{transition}} = 81.6636$ and $t_{\text{transition}\to\text{post}} = 96.3869$ are applied at $t = 81.67$ and $t = 96.67$ on the full-time output grid. After the second switch the ROM remains in the post-reversal model until $t = 300$. Thus, the prediction part of this experiment is not the identification of a new regime switch, but the continuation of the last trained active regime beyond its training horizon.

This choice is intentional. The last available local model is trained on the post-reversal regime, so a later switch back to a pre-reversal or transition model would not correspond to a newly learned future reversal. If the full FOM trajectory underwent another circulation reversal after $t = 150$, the present three-regime ROM would probably not capture it without additional local models and a switching rule trained or validated for such later events.

The main capability demonstrated by this experiment is stable post-training continuation. The mean relative state error on the prediction interval $t > 150$ is 0.18786, compared with 0.15288 on $t \leq 150$, and the final error at $t = 300$ is 0.19144. The maximum error after $t = 150$, 0.26538, is smaller than the full-run maximum 0.48122, because the largest error occurs during the reversal and switching transient rather than during the later post-reversal

Table 5.11: Prediction performance of the scheduled exact OpInf ROM compared with the full-time $t = 300$ FOM. The training-window comparison uses $t \leq 150$, while the post-training prediction interval uses $t > 150$.

Interval	Points	Mean e_x	Median e_x	Max. e_x	Final e_x	e_K^{RMS}	$e_{\ell_z}^{\mathrm{RMS}}$
Full run	871	0.17095	0.17574	0.48122	0.19144	0.14959	0.08318
$t \leq 150$	421	0.15288	0.13985	0.48122	0.23008	0.11093	0.08538
$t > 150$	450	0.18786	0.19008	0.26538	0.19144	0.18173	0.08128

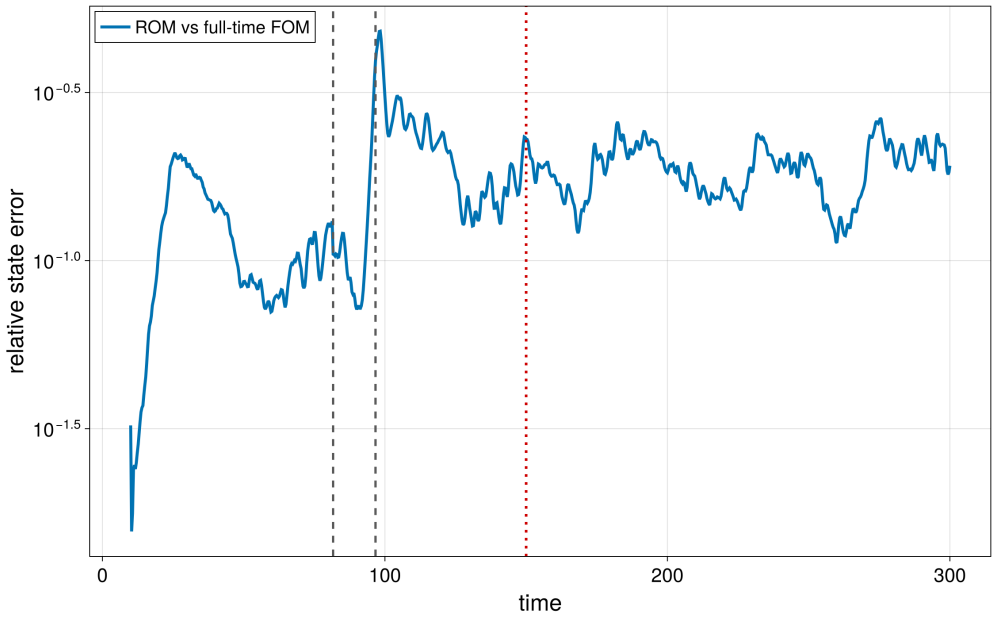


Figure 5.14: Relative state error for the scheduled exact OpInf ROM evaluated against the full-time $t = 300$ FOM. The grey dashed lines are the scheduled training-interval switch times and the red dotted line marks the end of the local training data at $t = 150$.

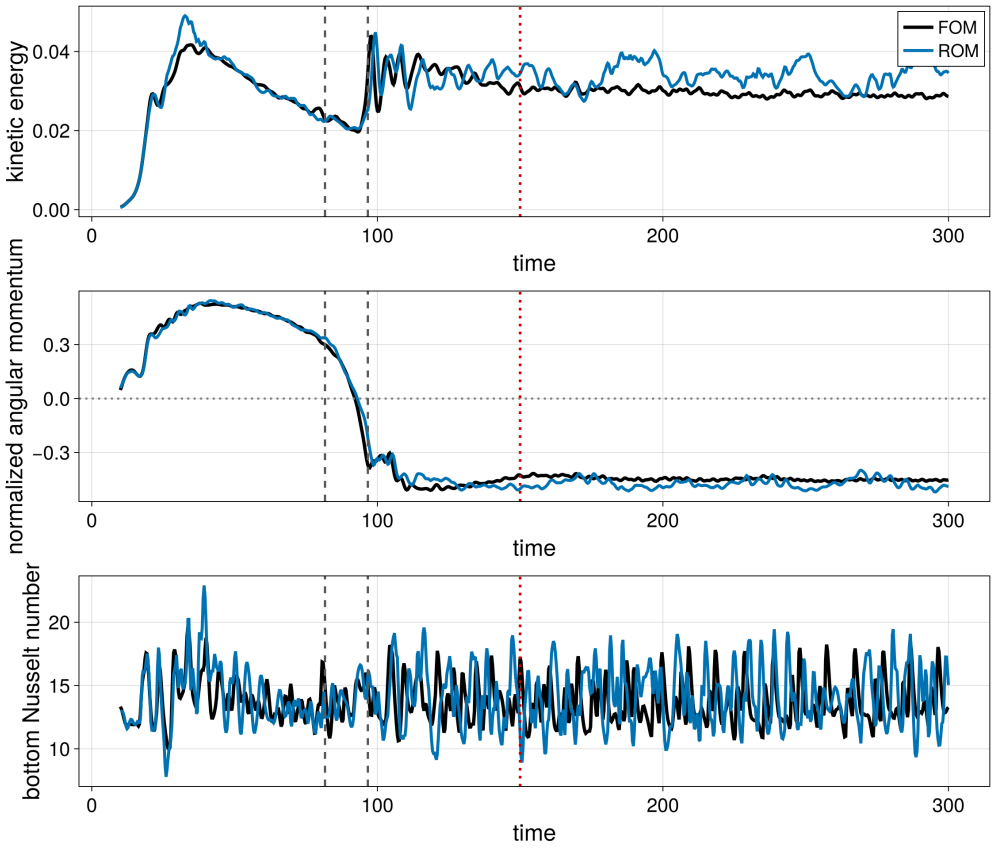


Figure 5.15: Prediction diagnostics for the scheduled exact OpInf ROM compared with the full-time FOM. The ROM remains stable after the training horizon and follows the post-reversal angular-momentum regime, but the kinetic energy is biased high and the heat-transfer diagnostic remains visibly noisy.

evolution. The normalized angular momentum is also predicted reasonably well after the training horizon: its RMS error is 0.08538 on $t \leq 150$ and 0.08128 on $t > 150$. This indicates that once the ROM has entered the post-reversal model, it can maintain the sign and approximate magnitude of the large-scale circulation for another 150 time units.

The same results also show the limits of the prediction. The kinetic-energy RMS error increases from 0.11093 before $t = 150$ to 0.18173 after $t = 150$, and the bottom-Nusselt RMS error increases from 0.16770 to 0.22404. Figure 5.15 shows the reason: the ROM follows the broad post-reversal state, but its kinetic energy remains slightly too large and more oscillatory than the FOM. The heat-transfer diagnostic is even more sensitive, because it depends on near-wall temperature gradients and small-scale thermal structures. These structures are precisely the components most easily affected by POD truncation and operator error. Therefore, the ROM is better at predicting the large-scale circulation state than at making quantitatively accurate long-horizon predictions of energetic and wall-gradient statistics.

This distinction is important for the interpretation of all Rayleigh–Bénard ROM results in this chapter. The scheduled switched ROMs demonstrate that exact OpInf can recover the intrusive projected dynamics when the relevant regimes and switch times are known. The $t = 300$ experiment keeps those training-derived switch times, so it should be interpreted as a scheduled continuation test rather than as a fully autonomous switching test. After the second switch the ROM simply evolves the learned post-reversal operator far beyond its training window. It can predict persistence of the learned post-reversal regime, but it has no additional local model for a later reversal or for a qualitatively new convective state. If the full FOM trajectory were to undergo another circulation reversal, the present ROM would either need a new switching policy and additional local models, or it would have to extrapolate outside the regimes represented in its training data.

The practical conclusion is that the exact OpInf switched ROM has useful predictive capability for medium-time continuation within a learned regime, especially for integral measures such as normalized angular momentum. For the aim of this thesis, this limitation is acceptable: the objective is to test whether exact non-intrusive operator inference can reproduce intrusive reduced operators and handle a reversal using regime-local ROMs. The prediction experiment strengthens that conclusion by showing that the $\eta = 0.99$ switched model remains stable and physically recognizable beyond the training interval, while also identifying the quantities for which extrapolation is least reliable.

5.3.6 Discussion

The Rayleigh–Bénard results support three main conclusions. First, exact operator inference works as intended for this two-dimensional incompressible buoyancy-driven test case. When the exact OpInf and intrusive ROMs use the same centered POD bases, the reduced operators agree to round-off and the switched trajectories are indistinguishable at the level of the reported statistics. This is a stronger result than merely obtaining a good trajectory fit: it verifies that the exact OpInf operator-query construction recovers the same sampled Galerkin-projected reduced model as the intrusive reference under the assumptions stated in Chapter 4. This supports the exact OpInf recovery result of Rosenberger et al. [29], now in the two-dimensional incompressible Rayleigh–Bénard setting considered here.

Second, Rayleigh–Bénard convection is sensitive to the retained fluctuation energy. The $\eta = 0.90$, 0.95, and 0.97 switched ROMs are fast, but they fail to reconstruct the diagnostics with sufficient accuracy. The $\eta = 0.99$ model is substantially more expensive, yet it is the only one in this sweep that captures the normalized angular momentum, kinetic energy and state trajectory at the level needed for the thesis objective. This reflects the physical nature of the problem: the reversal is not governed by one coherent roll alone, but by an interaction of thermal plumes, boundary-layer structures and the large-scale circulation. A basis that discards apparently small fluctuation-energy modes may still discard modes that are dynamically important during the transition.

Third, the ROM has a specific but limited predictive capability. In the $t = 300$ full-time comparison, the scheduled exact OpInf ROM remains stable after its training horizon and keeps a mean post-training state error of 0.18786. It therefore predicts persistence of the learned post-reversal regime reasonably well. However, the kinetic-energy and Nusselt errors grow after $t = 150$, and the model has no representation of later regimes that were absent from the training data. Its prediction capability is therefore best understood as medium-time scheduled continuation inside a learned regime, not as autonomous long-horizon forecasting of arbitrary future reversals.

Overall, the scheduled switched exact OpInf ROM provides the best balance for the purpose of this thesis. It demonstrates that the exact OpInf method can reproduce the sampled intrusive reduced operators for a two-dimensional incompressible thermal flow without explicitly assembling the full-order polynomial matrices, and that regime-local bases can handle a circulation reversal better than a single global description. Its main limitations are cost and extrapolation. Once $\eta = 0.99$ is required, the maximum local rank becomes 73, and the online reduced system is no longer extremely cheap. Once the trajectory leaves the regimes covered by the local models, predictive reliability is not guaranteed. The method should therefore be viewed as an accuracy-oriented data-driven ROM rather than as the lowest-cost surrogate or a complete replacement for the full model.

Chapter 6

Conclusions

This thesis studied data-driven reduced-order models (ROMs) for two-dimensional incompressible flows with coherent vortex and thermal-convection dynamics. The main objective was to assess whether exact operator inference can recover intrusive projection-based reduced operators without explicitly assembling the full-order polynomial operators, and whether such models can provide useful surrogates for vortex-driven and buoyancy-driven flow problems. Two test cases were considered: the normal dipole–wall collision at $Re = 2500$, and Rayleigh–Bénard convection at $Ra = 1.6 \times 10^7$ and $Pr = 1$.

The first conclusion is that the exact operator-inference construction was verified successfully for both incompressible-flow test cases. In the dipole–wall problem, exact OpInf recovered the matrix-free intrusive Galerkin operators to round-off accuracy, with relative operator errors of order 10^{-13} for the linear term and 10^{-14} for the quadratic term. The resulting trajectory was indistinguishable from the intrusive ROM trajectory at the reported precision. The same conclusion was obtained for the Rayleigh–Bénard switched ROM: for each local regime, the affine, linear and quadratic operators inferred by exact OpInf agreed with the corresponding sampled intrusive Galerkin operators to round-off accuracy. These results confirm that the rank-ensuring one-step query strategy can reproduce the sampled intrusive reduced model in the two-dimensional finite-volume setting used in this work.

The second conclusion is that exact recovery of the intrusive ROM should not be confused with exact recovery of the full-order solution. In the dipole–wall test case, the 1536×1536 FOM reproduced the benchmark enstrophy peaks and near-wall vorticity structure, so it provided a validated reference for the ROM study. The exact OpInf and intrusive Galerkin ROMs followed the FOM kinetic-energy decay well, but the global POD basis struggled after the primary wall collision, where thin vorticity layers and localized secondary vortices dominate the error. This shows that exact OpInf removes the operator-identification error relative to the intrusive ROM, but it does not remove the truncation and closure limitations of a global projection basis. For flows whose important structures are strongly localized, improving the reduced space is at least as important as improving the operator fit.

The third conclusion is that trajectory-based operator inference and exact operator inference serve different roles. Conventional trajectory OpInf can be much cheaper online when the available snapshots force a lower identifiable rank, and in some low-rank comparisons it behaves like a closure-corrected model for the retained coordinates. However, this lower cost comes with a change in what is being compared. In the dipole case, the trajectory OpInf model had to be reduced to rank 8, whereas the $\eta = 0.99$ exact OpInf model used rank 22. In the Rayleigh–Bénard method comparison, trajectory OpInf used ranks $30/15/26$, whereas the $\eta = 0.99$ exact OpInf and intrusive switched ROMs required ranks $72/22/73$. The trajectory-based models were therefore faster partly because they advanced smaller systems, not only because of the inference method. Exact OpInf is the appropriate tool when the goal is to recover the intrusive projected dynamics in a known polynomial feature space; trajectory OpInf is more naturally interpreted as a cheaper empirical surrogate

whose accuracy depends on the available trajectory data and regularization choices.

The Rayleigh–Bénard experiments also showed that retained fluctuation energy is a critical modelling choice. Because the mean thermal structure contains a large part of the raw state energy, selecting POD ranks from the uncentered energy alone can hide dynamically important velocity and temperature fluctuations. The scheduled switched ROMs based on $\eta = 0.90$, $\eta = 0.95$, and $\eta = 0.97$ were computationally attractive but did not reproduce the reversal diagnostics reliably. The $\eta = 0.99$ model was much more expensive, but it was the only one in the sweep that accurately followed the state trajectory, kinetic energy and normalized angular momentum through the circulation reversal. Thus, for this problem, a high fluctuation-energy threshold was necessary to represent the coupled plume, boundary-layer and large-scale-roll dynamics involved in the reversal.

The regime-local switched ROM was useful for the Rayleigh–Bénard trajectory with a circulation reversal. By training separate local models before, during and after the reversal, and by switching between them using times derived from the FOM angular-momentum signal, the ROM could represent dynamics that would be difficult for one global basis of comparable size. The scheduled prediction experiment extended this result beyond pure reconstruction: after being trained only up to $t = 150$, the exact OpInf switched ROM remained stable up to $t = 300$ and reproduced the persistence of the post-reversal circulation with a mean post-training state error of 0.18786. Nevertheless, the kinetic-energy and Nusselt errors increased after the training horizon, showing that the model is more reliable for large-scale circulation measures than for small-scale or wall-gradient-sensitive diagnostics.

Several limitations follow from these results. First, the 256×256 Rayleigh–Bénard trajectory used for the switched-ROM study should be interpreted as a selected FOM trajectory for reduced-modelling experiments, not as a grid-converged prediction of the physical Rayleigh–Bénard flow at this Rayleigh number. The finer-grid runs showed that the large-scale regime and heat-transfer diagnostics are resolution-sensitive. Second, the scheduled switched ROM uses FOM-derived switching times, so it is not a fully autonomous online decision procedure. The $t = 300$ prediction test should therefore be read as continuation of the learned post-reversal regime, not as a test of whether the ROM can discover future regime changes. A later circulation reversal would require additional local models and a switching rule trained or validated for such events. Third, the high-rank $\eta = 0.99$ Rayleigh–Bénard ROM was only moderately faster than the FOM in the present implementation, because no hyper-reduction or specialized online optimization was used and because the local quadratic operators become costly at ranks above 70.

Overall, the work supports a clear but qualified conclusion. Exact operator inference is a reliable way to reproduce intrusive affine-quadratic reduced operators from full-order model queries in the tested two-dimensional incompressible-flow settings. It is particularly valuable as an accuracy-oriented data-driven alternative when explicit full-order operator assembly is impractical. Its performance as a practical surrogate, however, is controlled by the reduced space, the amount of dynamically relevant fluctuation energy retained, and the regime coverage of the training data. Future work should therefore focus on improving the reduced representation, online efficiency and physical resolution near dynamically important regions. This includes space-local POD for wall layers and plumes, adaptive or state-dependent switching rules, hyper-reduction of the online nonlinear terms, stretched Rayleigh–Bénard grids with finer cells near the walls, and further tests on grid-converged trajectories and longer sequences with multiple reversals. These possible extensions are discussed in more detail in Appendix C.

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Appendix A

Intrusive assembly and exact operator-inference checks

The two sections in this appendix describe the same conceptual construction: an intrusive ROM is obtained by evaluating the implemented semi-discrete FOM right-hand side on states in the POD space and projecting the result with the discrete M-inner product. The difference is practical rather than conceptual. For the dipole–wall case, the velocity-only operator is assembled directly from projected diffusion and convection evaluations. For the Rayleigh–Bénard case, the coupled velocity–temperature operator is assembled through projected samples of the full right-hand side, which is the same sampled Galerkin form used to check exact operator inference.

A.1 Dipole–wall intrusive assembly

For the dipole–wall collision test case, the full-order state $\boldsymbol{x}(t) \in \mathbb{R}^N$ contains only the discrete velocity degrees of freedom. The velocity boundary conditions are homogeneous no-slip conditions and no affine forcing term is introduced in the polynomial ROM. Hence, for this case one may write the semi-discrete velocity equation abstractly as

$$\dot{\boldsymbol{x}}(t) = A_1 \boldsymbol{x}(t) + A_2 \boldsymbol{x}^2(t),$$

or, equivalently,

$$\dot{\boldsymbol{x}}(t) = \mathscr{P}_h \mathscr{D}_h(\boldsymbol{x}(t)) + \mathscr{P}_h \mathscr{C}_h(\boldsymbol{x}(t)).$$

Here, $\mathscr{D}_h$ denotes the discrete viscous contribution, $\mathscr{C}_h$ denotes the discrete convective contribution, and $\mathscr{P}_h$ denotes the discrete pressure projection used to enforce incompressibility. The notation $\mathscr{C}_h(\boldsymbol{x})$ represents the quadratic convection operator evaluated at the velocity field $\boldsymbol{x}$.

Let

$$\boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\phi}_1 & \cdots & \boldsymbol{\phi}_r \end{bmatrix} \in \mathbb{R}^{N \times r}$$

be the M-orthonormal POD basis. Instead of assembling the full-order matrices A_1 and A_2, the implementation constructs the reduced operators directly. For each POD mode $\boldsymbol{\phi}_j$, the linear column is obtained by applying the full-order diffusion routine to the corresponding full-order velocity field, applying the same pressure projection as in the FOM, and projecting the result onto the POD space:

$$\widetilde{A}_{1,:,j} = \boldsymbol{\Phi}^T M \mathscr{P}_h \mathscr{D}_h(\boldsymbol{\phi}_j), \qquad j = 1, \ldots, r.$$

The homogeneous velocity boundary conditions are applied to the reconstructed velocity field before the full-order operator is evaluated, so that the reduced assembly uses the same boundary treatment as the FOM.

The quadratic operator is assembled in a matrix-free way by using the quadratic structure of the convective term. First, the diagonal quadratic contributions are computed as

$$\boldsymbol{b}_{jj} = \boldsymbol{\Phi}^T M \mathscr{P}_h \mathscr{C}_h(\boldsymbol{\phi}_j), \qquad j = 1, \ldots, r.$$

For the mixed products, the implementation evaluates the convection operator at sums of POD modes and uses the polarization identity

$$\boldsymbol{b}_{jk} = \boldsymbol{\Phi}^T M \mathscr{P}_h \left[\mathscr{C}_h(\boldsymbol{\phi}_j + \boldsymbol{\phi}_k) - \mathscr{C}_h(\boldsymbol{\phi}_j) - \mathscr{C}_h(\boldsymbol{\phi}_k) \right], \qquad 1 \le j < k \le r.$$

Thus, if $\boldsymbol{a}^2$ stores the unique products

$$\boldsymbol{a}^2 = (a_1^2,\ a_1 a_2, \ldots, a_1 a_r,\ a_2^2, \ldots, a_r^2)^T,$$

then the columns of $\widetilde{A}_2$ are precisely the vectors $\boldsymbol{b}_{jk}$, ordered according to the corresponding product $a_j a_k$. The resulting reduced system is therefore

$$\dot{\boldsymbol{a}}(t) = \widetilde{A}_1 \boldsymbol{a}(t) + \widetilde{A}_2 \boldsymbol{a}^2(t),$$

with no affine term for the dipole–wall case.

Equivalently, the implementation stores the same quadratic information in a symmetric third-order tensor $\widetilde{Q}$. In that representation,

$$\widetilde{Q}_{:,j,j} = \boldsymbol{b}_{jj}, \qquad \widetilde{Q}_{:,j,k} = \widetilde{Q}_{:,k,j} = \frac{1}{2}\boldsymbol{b}_{jk}, \qquad j < k,$$

so that

$$\sum_{j=1}^{r}\sum_{k=1}^{r} \widetilde{Q}_{:,j,k} a_j a_k = \widetilde{A}_2 \boldsymbol{a}^2.$$

This procedure is intrusive because it calls the internal full-order routines for diffusion, convection, boundary conditions and pressure projection. However, it remains matrix-free: neither A_1, A_2, nor the full convection tensor is materialized.

A.2 Rayleigh–Bénard sampled intrusive assembly and exact OpInf check

The operator comparison in Table 5.7 uses the same Galerkin idea as Appendix A.1, but the Rayleigh–Bénard implementation forms the intrusive operators from projected samples rather than by listing every reduced tensor entry explicitly. The intrusive ROM is therefore not an independently assembled continuous weak-form model. It is the sampled Galerkin projection of the implemented semi-discrete Boussinesq right-hand side, evaluated with the same boundary treatment, pressure projection and velocity–temperature state representation as the FOM.

Let the local centered POD approximation be

$$\boldsymbol{x} = \boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a},$$

where $\boldsymbol{x}$ contains both velocity and temperature degrees of freedom. For a reduced sample $\boldsymbol{a}$, the intrusive derivative sample is

$$\boldsymbol{g}_{\mathrm{int}}(\boldsymbol{a}) = \boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{f}(\boldsymbol{\mu} + \boldsymbol{\Phi}\boldsymbol{a}),$$

where f denotes the implemented semi-discrete FOM right-hand side after applying the same projection and boundary routines as in the full model. The exact OpInf construction uses the same lifted reduced state, advances it by one explicit-Euler-consistent query,

$$\boldsymbol{x}_1 = \boldsymbol{x}_0 + \Delta t\, \boldsymbol{f}(\boldsymbol{x}_0),$$

and forms

$$\boldsymbol{g}_{\text{eOpInf}}(\boldsymbol{a}) = \boldsymbol{\Phi}^T \mathbf{M} \frac{\boldsymbol{x}_1 - \boldsymbol{x}_0}{\Delta t}.$$

Therefore

$$\boldsymbol{g}_{\text{eOpInf}}(\boldsymbol{a}) = \boldsymbol{\Phi}^T \mathbf{M} \boldsymbol{f}(\boldsymbol{x}_0) = \boldsymbol{g}_{\text{int}}(\boldsymbol{a})$$

up to floating-point round-off. Since the rank-ensuring reduced samples make the operator inference feature matrix nonsingular, the unregularized exact OpInf solve identifies the same affine-quadratic reduced map as this sampled intrusive projection.

This explains why the Rayleigh–Bénard exact OpInf and intrusive scheduled ROMs produce matching operators and matching trajectories in the reported results. It is an implementation consistency check: both procedures query the same semi-discrete right-hand side on the same reduced samples. The conclusion would not automatically hold for a trajectory-only operator-inference fit, for derivatives estimated by finite differences from stored snapshots, or for one-step data generated by a time integrator that is not consistent with the derivative quotient used in the inference system.

Appendix B

Effect of the optional viscous dissipation term

The full-order solver used in this work contains an optional viscous-heating term in the temperature equation. This term is not part of the classical Oberbeck–Boussinesq formulation commonly used for Rayleigh–Bénard convection, where the temperature equation contains advection and thermal diffusion but no volumetric heat source. In the notation of the present code, the equations can be written schematically as

$$\nabla \cdot \boldsymbol{u} = 0, \tag{B.1}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + \nu \nabla^2 \boldsymbol{u} + g\, \theta\, \boldsymbol{e}_g, \tag{B.2}$$

$$\frac{\partial \theta}{\partial t} + \boldsymbol{u} \cdot \nabla \theta = \kappa \nabla^2 \theta + \Phi_\nu. \tag{B.3}$$

When the option dodissipation is disabled, $\Phi_\nu = 0$. When it is enabled, the implementation adds

$$\Phi_\nu = \nu \|\nabla \boldsymbol{u}\|_F^2, \tag{B.4}$$

where $\|\cdot\|_F$ denotes the Frobenius norm of the velocity-gradient tensor. This is a non-negative source term and therefore represents internal heating of the fluid. The inclusion of such a term is motivated by the thermodynamic energy balance: kinetic energy removed by viscous stresses may be returned to the internal-energy equation as heat [33].

This mechanism can be useful when the objective is to study energy-consistent discretizations or flows in which viscous heating is physically relevant. Sanderse and Trias [33], for example, show that adding a properly constructed viscous-dissipation term can give a discrete balance in which the kinetic-energy sink due to viscosity is exactly matched by a thermal source. Viscous dissipation is also of interest in thermal convection because it is spatially intermittent and can be concentrated differently in the bulk and in boundary layers [6]. In such settings, the term is not merely a numerical detail: it changes the thermal energy budget of the flow.

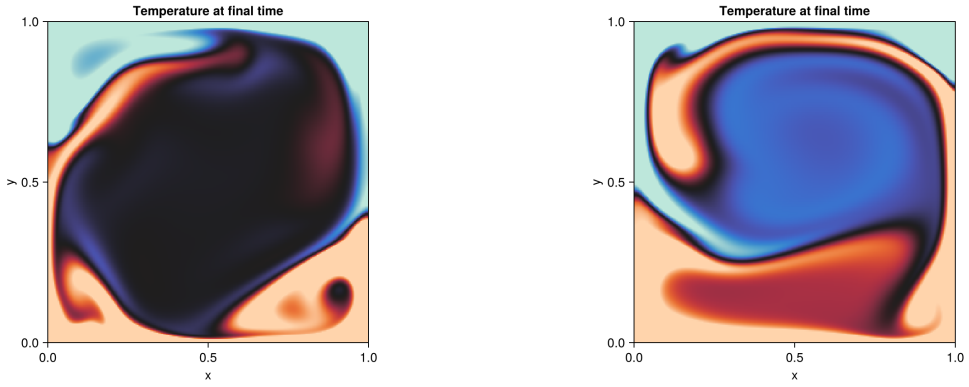
However, the Rayleigh–Bénard experiments in this thesis are intended to represent the standard Oberbeck–Boussinesq problem, where the imposed plate temperature difference is the only thermal driving. The temperature scale is $\Delta T = T_h - T_c$, and the Nusselt number measures heat transport relative to conduction between the plates [2]. Adding (B.4) changes this physical problem by introducing volumetric heat generation inside the domain. As a result, the heat entering and leaving through the two plates need not be equal. The mean temperature budget can be written as

$$\frac{d\langle\theta\rangle}{dt} = \frac{\kappa \Delta T}{H^2}\,(Nu_b - Nu_t) + \langle \Phi_\nu \rangle, \tag{B.5}$$

so that at statistical steady state a positive internal heat source implies

$$Nu_t - Nu_b = \frac{H^2}{\kappa \Delta T} \langle \Phi_v \rangle. \tag{B.6}$$

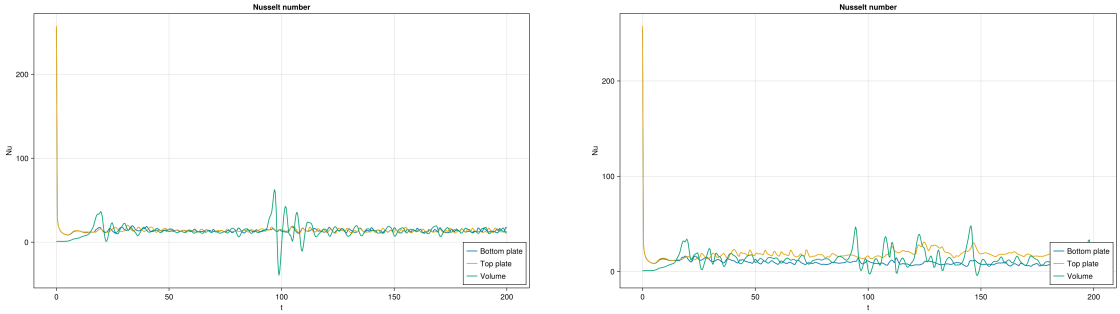
This imbalance between the two plate Nusselt numbers is consistent with observations reported for Rayleigh–Bénard simulations with viscous heating [33]. It explains why enabling `dodissipation` can produce a visible change in the Nusselt numbers and in the spatial temperature field, even when the viscosity is numerically small. For this reason, all experiments reported in the main text set `dodissipation = false`. The comparisons in Fig. B.2 and Fig. B.1 are included only to document the effect of this modelling choice.



(a) Without viscous dissipation, `dodissipation = false`.

(b) With viscous dissipation, `dodissipation = true`.

Figure B.1: Final temperature field for the Rayleigh–Bénard simulation with and without the optional viscous-dissipation source term. When `dodissipation = true`, kinetic energy dissipated by viscosity is added as a positive volumetric heat source in the temperature equation. This modifies the thermal balance and visibly changes the temperature distribution in the domain. The simulations shown here are used only to illustrate the effect of this modelling option; the experiments in the main text use `dodissipation = false`.



(a) Without viscous dissipation, do`dissipation = false`.

(b) With viscous dissipation, do`dissipation = true`.

Figure B.2: Nusselt-number histories for the same Rayleigh–Bénard configuration with and without the optional viscous-dissipation term. In the classical Oberbeck–Boussinesq case, heat is supplied and removed only through the plates. With viscous dissipation enabled, heat is also generated inside the fluid, so the bottom and top plate Nusselt numbers need not balance in the same way. This explains the change in heat-transfer diagnostics observed when do`dissipation` is enabled.

Appendix C

Further possible developments

The main text focuses on global POD bases and, for some Rayleigh–Bénard runs, on switching between several global local bases according to the FOM-derived angular-momentum schedule described in Section 4.4. Another possible direction, not used in the reported numerical experiments, is to replace a single global POD basis by a space-local POD construction. This would allow different parts of the domain to have different local resolution, which may be useful for flows with different spatial structures such as the dipole–wall collision test case.

C.1 Space-local POD with multiple local bases

The standard POD construction in Chapter 4 produces a single global basis. Each retained mode may have support throughout the whole spatial domain. This is optimal for the global projection error in the chosen weighted norm, but it can be inefficient when the relevant flow structures are spatially localized or when different parts of the domain require different local resolution. Space-local POD addresses this by computing several local POD bases on spatial blocks and combining them into one reduced representation [3].

Let the state degrees of freedom be partitioned into n_b non-overlapping spatial blocks

$$\mathscr{I}_1, \mathscr{I}_2, \ldots, \mathscr{I}_{n_b}, \qquad \mathscr{I}_i \cap \mathscr{I}_j = \emptyset \quad (i \neq j).$$

If $\mathbf{R}_\ell$ denotes the row-selection matrix for block $\mathscr{I}_\ell$, then the corresponding local snapshot matrix and local weight matrix are

$$\mathbf{X}_\ell = \mathbf{R}_\ell \mathbf{X}, \qquad \mathbf{M}_\ell = \mathbf{R}_\ell \mathbf{M} \mathbf{R}_\ell^T.$$

A weighted POD is then computed independently in each block,

$$\mathbf{M}_\ell^{1/2} \mathbf{X}_\ell = \mathbf{U}_\ell \mathbf{\Sigma}_\ell \mathbf{W}_\ell^T,$$

and the first r_ℓ local modes define a local basis $\mathbf{\Phi}_\ell \in \mathbb{R}^{|\mathscr{I}_\ell| \times r_\ell}$ satisfying

$$\mathbf{\Phi}_\ell^T \mathbf{M}_\ell \mathbf{\Phi}_\ell = \mathbf{I}_{r_\ell}.$$

After inserting each local basis into its corresponding rows of the full state vector, the global space-local basis has a block-sparse structure,

$$\mathbf{\Phi}_{\mathrm{SL}} = \begin{bmatrix} \mathbf{\Phi}_1 & & \\ & \ddots & \\ & & \mathbf{\Phi}_{n_b} \end{bmatrix}, \qquad r_{\mathrm{SL}} = \sum_{\ell=1}^{n_b} r_\ell. \tag{C.1}$$

This expression should be understood up to the ordering of the degrees of freedom. In an implementation with the original state ordering, the local modes are inserted into the rows belonging to their spatial block, so the resulting matrix is sparse and has nonzero entries only inside the selected block.

Although several local bases are used, the result is still a single reduced state vector. The projection and reconstruction formulas remain the same as in (4.7)–(4.8), with $\boldsymbol{\Phi}$ replaced by $\boldsymbol{\Phi}_{\mathrm{SL}}$. Thus,

$$\boldsymbol{x}_r(t) = \boldsymbol{\mu} + \boldsymbol{\Phi}_{\mathrm{SL}}\boldsymbol{a}(t), \qquad \boldsymbol{a}(t) = \boldsymbol{\Phi}_{\mathrm{SL}}^T\mathbf{M}(\boldsymbol{x}(t) - \boldsymbol{\mu}).$$

This is different from the switched ROM in Section 4.4. A switched ROM uses different global bases at different scheduled times associated with different trajectory regimes. Space-local POD instead uses several local spatial bases simultaneously inside one fixed reduced space.

The local ranks r_ℓ may be prescribed explicitly, or they may be determined from the local singular values. In a per-block energy strategy, each block chooses the smallest r_ℓ such that

$$\frac{\sum_{i=1}^{r_\ell}\sigma_{\ell,i}^2}{\sum_i \sigma_{\ell,i}^2} \geq \eta_{\mathrm{loc}},$$

so the total dimension r_{SL} is an outcome of the local thresholds. In a fixed-rank strategy, the singular values from all blocks are pooled and the largest r_{SL} values are retained. This allocates more local modes to blocks with more energetic snapshot content while keeping the total reduced dimension fixed.

For the dipole–wall problem, such a construction could be useful because the most demanding structures are localized near the wall after impact. For Rayleigh–Bénard convection, it could be used to assign different local resolution to boundary layers, plume regions and the interior of the cell. Since this strategy was not used in the experiments reported in this thesis, assessing its effect on projection error, stability and exact operator inference accuracy is left as future work.

C.2 Additional directions

Several other extensions follow directly from the limitations observed in Chapter 5. First, the scheduled switched ROM could be replaced by a more autonomous switching rule. In the main Rayleigh–Bénard reconstruction experiments, the switching times are derived from the FOM angular-momentum history and then imposed on the ROM. A more predictive strategy would decide when to switch using only ROM-available quantities, such as reconstructed angular momentum, local projection errors, or learned classifiers for the active regime.

Second, the online cost of the high-rank Rayleigh–Bénard ROMs could be reduced by hyper-reduction. In a nonlinear ROM, a small reduced state does not automatically guarantee a cheap online method: nonlinear fluxes, projected residuals, reconstruction-based diagnostics, or high-rank quadratic operators can still be expensive. Hyper-reduction refers to approximating these nonlinear terms from a small set of sampled cells, interpolation points, or reduced residual bases, so that the online cost no longer scales like a full-order evaluation. Such techniques would be especially relevant for the $\eta = 0.99$ Rayleigh–Bénard ROM, where the local ranks reach $72/22/73$ and the reduced quadratic operators dominate the online cost. Relevant examples include the discrete empirical interpolation method for nonlinear terms [12], the GNAT sample-mesh approach for CFD-type reduced models [9], and mesh sampling and weighting for nonlinear Petrov–Galerkin ROMs with local reduced bases [17].

Third, future Rayleigh–Bénard FOM data could use stretched grids. The present dipole–wall study already uses grid stretching to cluster points near the right wall, where thin vorticity layers are generated after impact. A similar idea could be useful for Rayleigh–Bénard convection by clustering cells near the hot and cold plates, where thermal and velocity boundary

layers determine the Nusselt numbers. This would provide a more targeted resolution of the wall-gradient-sensitive diagnostics and should be combined with a new grid-convergence study before retraining the ROMs.

Finally, longer Rayleigh–Bénard trajectories with several circulation reversals would require either additional local models or a more flexible regime-discovery strategy. The $t = 300$ prediction test in the main text shows stable continuation inside the learned post-reversal regime, but it does not test whether the same ROM can discover and represent later regimes that were absent from the training data.